MIL-M-49234A(CR)
20 April 1987
SUPERSEDING
MIL-M-49234
17 November 1981

MILITARY SPECIFICATION

MODEM, DIGITAL DATA, MD-1026(P)/G

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the construction and testing necessary for the manufacture of Modem, Digital Data, MD-1026(P)/G. This unit is also known as Group Modem (GM) and is one element of the Digital Group Multiplexer (DGM) family of cable system equipments developed by the Joint Tactical Command, Control and Communications Agency (JTC^3A, formerly TRI-TAC). See 6.1 for intended use of the MD-1026(P)/G.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communication-Electronics Command, ATTN: AMSEL-ED-LC, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 5820

Distribution Statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-C-55583	Cable Assembly, Special Purpose CX-11230/G, Cable Assembly Adapter CS-10734/G
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirement for Electronic Equipment
MIL-STD-461 Notice 4	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics Measurement of
MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-2073-2	Packaging Requirements Code
MIL-STD-45662	Calibration System Requirements

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of the solicitation.

SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments
CR-CX-0174-001	TEMPEST Provisions for the Family of Digital Group Multiplexer Equipments

DRAWINGS

Communications-Electronics Command

DL-SM-B-875932	Modem,Digital Data, MD-1026(P)/G
SM-D-875948	Circuit Card Assembly, BITE 1
SM-D-875969	CCA, Conditioned Diphase A
SM-D-875971	CCA, Dipulse
SM-D-875992	CCA, Cable Orderwire
SM-D-875991	CCA, Cable Orderwire
SM-D-875992	CCA, Conditioned Diphase B
SM-D-884567	CCA, Multi Vdc
SM-D-884579	CCA, AC Input
SM-D-884593	CCA, 5 Vdc

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets, or MS standards) the text of this specification shall take precedence. Nothing in this specificatioin, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with requirements of this specification and of DL-SM-B-875932 including the requirements for parts, materials and processes therein.

3.2.2 Printed wiring. The printed wiring shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirement 30 and 64.

3.3 Parts, materials, and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and drawing DL-SM-B-875932.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in according with requirements specified in the contract or purchase order.

3.5.1 Serial numbers (see 4.15). A unique serial number is required on each nomenclatured item and Circuit Card Assembly (CCA). The numbers shall be consecutively assigned.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled MD-1026(P)/G shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as: (1) no built-in test equipment (BITE) alarms during burn-in, and (2) the successful completion of the operational test of 4.7.8 after burn-in.

3.7 Electrical. (See 4.7.1)

3.7.1 Balanced equipment-side signal characteristics.

3.7.1.1 Line-to-line voltage levels. (See 4.7.1.1) The line-to-line voltage shall be between +0.1 and +4.0 volts for a digital 1 and between -0.1 and -4.0 for a digital 0, measured into a 100 ohm dummy load. For the 2 kb/s data and timing outputs, the minimum voltage when measured into a 100 ohm dummy load shall be +0.5 volts for a digital 1 and -0.5 volts for a digital 0.

3.7.1.2 Line-to-ground voltage. (See 4.7.1.2) The magnitude of the line-to-ground voltage shall not exceed 3 volts.

3.7.1.3 Signal balance. (See 4.7.1.3) The magnitude of the difference between the magnitudes of the digital 1 level and the digital 0 level shall be less than 10% of the larger of the two.

3.7.1.4 Receiver sensitivity. (See 4.7.1.4) The receiver shall deliver the proper digital output when a line-to-line voltage of +50mV for a digital 1 or -50mV for a digital 0 is applied to the receiver input terminals.

3.7.1.5 Clockwidth tolerance. (See 4.7.1.5) The positive half of the clock waveform shall be equal to half the period $\pm$15 ns for group rate timing signals and $\pm$1500 ns for 2 and 16 kb/s timing signals.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currenversion before use.

3.7.1.6 Clock/data phasing. (See 4.7.1.6) For those data signals having a related clock signal, the time difference between the upward transition of the clock signal and either the upward or downward transition of the data signal shall be no more than 29 ns for group rate signals, or 1500 ns for 2 and 16 kb/s signals. Times shall be measured at the zero voltage point of the signals.

3.7.1.7 Offset. (See 4.7.1.7) The magnitude of the offset voltage measured between the center tap of a 100 ohm dummy load and ground shall not be greater than 3 volts.

3.7.1.8 Slew rate. (See 4.7.1.8) When 100 feet of RG-108A/U cable terminated in a 78 ohm dummy load is connected to the driver output, the transition betweeen ±40 mV (either direction) shall take no more than 10 ns for group rate signals and 100 ns for the 2 and 16 kb/s signals.

3.7.1.9 Input load impedance. (See 4.7.1.9) The input impedance shall be 100 ohms ±10%.

3.7.1.10 Unloaded output. (See 4.7.1.10) For the 2 kb/s data and timing outputs, the maximum voltage into an open circuit shall be +3.0 volts for a digital 1 and -3.0 volts for a digital 0.

3.7.2 Cable-side group signal characteristics. (See 4.7.2) There are three types of modules used in the GM case: Diphase, Dipulse, and Bipolar.

3.7.2.1 Output voltage levels: diphase, dipulse, and bipolar. (See 4.7.2.1)

a. Diphase signal output voltage level. The output voltage level into a 58-Ohm resistive dummy load shall be 3.0V peak-to-peak ±20%.

b. Dipulse signal output voltage level and pulse width. The output voltage level into a 58-Ohm resistive dummy load shall be from 1.6 to 2.0V base to peak. The pulse width shall be from 180 to 260 ns, measured at the 50% points of the rise and fall slopes.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

c. Bipolar signal output voltage level and pulse width. The output voltage level into a 59-Ohm resistive dummy load shall be from 0.9±0.1V base to level. The pulse width shall be 90 to 110 ns measured at the 50% amplitude points.

3.7.2.2 Output rise and fall time. (See 4.7.2.2)

a. Diphase signal. The rise and fall times measured from 10 to 90% into 58 ohms shall be less than 30 ns for group rates above 576 kbps and up to 130 ns for group rates 526 kbps and below.

b. Dipulse signal. The rise time shall be 40 ns and the fall time shall be 50 ns measured into 58 ohms.

c. Bipolar signal. The rise and fall time shall be 25 ns when measured into 58 ohms.

3.7.2.3 Output impedance, diphase and bipolar signals. (See 4.7.2.3)

a. Diphase signal. The output impedance shall be 58 ohms ±15% for data rates 72 to 4608 kbps.

b. Bipolar signal. The output impedance shall be between 42 to 75 ohms from 400 KHz to 3 MHz.

3.7.2.4 Input impedance, diphase and bipolar signals. (See 4.7.2.4)

3.7.2.4.1 Diphase signals. (See 4.7.2.4.1) The input impedance shall be 50 to 100 ohms from 300 to 1000 Hz, and 58 ohms ±15% from 1000 Hz to 3000 kHz.

3.7.2.4.2 Bipolar signals. (See 4.7.2.4.2) The input impedance shall be 42 to 75 ohms from 400 kHz to 3 MHz.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.7.2.5 Asymmetry. (See 4.7.2.5).

3.7.2.5.1 Transmit diphase asymmetry. (See 4.7.2.5.1) The displacement of the center zero crossing of a conditioned diphase "zero" bit from the ideal center of the bit shall not be more than ±5% of the total bit period.

3.7.2.5.2 Receiver diphase asymmetry. (See 4.7.2.5.2) The diphase receiver shall operate error free at each bit rate when a signal transmitted with ±5% asymmetry is received over the specified range of cable lengths. The test shall be performed with combined digital orderwire signal present at rates of 256 kb/s and higher and with the analog voice orderwire (AVOW) signal present at rates of 144 kb/s and lower.

3.7.3 Orderwire signal characteristics. (See 4.7.3) The diphase module shall provide a combined digital orderwire and a maintenance orderwire (MOW). Only one of these can be used at a time. The dipulse and bipolar modules have a maintenance orderwire only.

3.7.3.1 Diphase maintenance orderwire. (See 4.7.3.1)

3.7.3.1.1 Cable side output. (See 4.7.3.1.1) With -4 dBm at 1 kHz applied to the equipment side input, the output into a 58-ohm resistive load shall be +5 dBm ±1 dB.

3.7.3.1.2 Equipment side output. (See 4.7.3.1.2) Over a range of -18.5 to +8 dBm at 1 kHz applied at the cable side input, the output shall be -7 to -2 dBm into a 600 ohm load.

3.7.3.1.3 Input impedance. (See 4.7.3.1.3) The input impedance shall be from 540 to 660 ohms over the frequency range from 300 Hz to 3400 Hz.

3.7.3.1.4 Signal-to-distortion ratio. (See 4.7.3.1.4) The signal-to-distortion ratio shall be 30 dB minimum at 1 KHz.

3.7.3.1.5 Output impedance. (See 4.7.3.1.5) The output impedance shall be 540 to 660 ohms over the frequency range from 300 Hz to 3400 Hz.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.7.3.2 Dipulse and bipolar maintenance orderwire. (See 4.7.3.2)

3.7.3.2.1 Cable-side output. (See 4.7.3.2.1) With -4 dBm at 1 kHz applied at the equipment side input, the output voltage shall be 2.75 Vrms ±10% into an 880-ohm load.

3.7.3.2.2 Equipment side output. (See 4.7.3.2.2) With an input on the cable side from 0.9 to 9.0 Vrms, for a dipulse modem or 1.5 to 3.0 Vrms for bipolar modem, the output shall be from -7 to -2 dBm into a 600 ohm load.

3.7.3.2.3 Cable-side input impedance, dipulse and bipolar. (See 4.7.3.2.3) The input impedance shall be 790 to 1390 ohms from 300 to 1700 Hz including the loading coil.

3.7.3.2.4 Cable-side output impedance, dipulse and bipolar. (See 4.7.3.2.4) The output impedance shall be 790 to 1390 ohms from 300 to 1700 Hz including the loading coil.

3.7.3.3 Combined digital orderwire cable-side output. (See 4.7.3.3) The output voltage into a 58-ohm resistive load shall be +0.5 V ±10% for a digital 1 and -0.5V ±10% for a digital 0.

3.7.3.4 Frequency response. (See 4.7.3.4) The loss vs frequency characteristics of the maintenance orderwire shall be as follows:

Frequency, Hz	Dipulse, Bipolar	Diphase
300	-3 to +6 dB	-5 to +8 dB
400	----------	±5 dB
1000	Reference	Reference
1700	±3 dB	---------
3000	----------	±5 dB
3400	----------	-5 to +8 dB

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

3.7.4 Power loop and current protection. (See 4.7.4) The dipulse and bipolar modules shall be capable of looping back up to 45mA of dc power feed current with a drop of no more than 50 volts. The cable diphase, dipulse and bipolar interface circuits shall withstand without damage to the equipment an injected current of 90 milliamperes on either center conductor of the CX-11230 cable (see MIL-C-55883).

3.7.5 Full duplex signal detection and dynamic range. (See 4.7.5)

3.7.5.1 Conditioned diphase. (See 4.7.5.1) The conditioned diphase module receiver shall recover the diphase signal from the cable with no errors when operating at 4608 kb/s over 0 to ½ mile of cable, at 2304 kb/s over 0 to 1 mile of cable, at 576 kb/s over 0 to 2 miles of cable, at 256 kb/s over 0 to 2 miles of cable, and 72 kb/s over 0 to 2 miles of cable. The error rate shall not be greater than that specified with the combined digital orderwire (rates above 144 kb/s) or with the ring signal of the maintenance orderwire being transmitted at the highest level, with multiplexer frame request acknowledge being transmitted and with the diphase signal attenuated 4 dB from the nominal level at the maximum cable length and with zero cable length and no attenuator. The conditioned diphase module receiver shall recover the diphase signal from the cable with no errors when the data rate is varied ± 30 PPM from the nominal. The digital voice orderwire (DVOW) and data orderwire (DOW) shall operate error free when the station clock signal is varied ± 30 PPM from the nominal.

3.7.5.2 Dipulse. (See 4.7.5.2) The dipulse module receiver shall recover the dipulse signal from the cable at 2304 kb/s +68 b/s with no errors when the signal is transmitted over 1/4, 1/2, 3/4, and 1 mile of cable. The error rate shall not be greater than that specified with the orderwire ring signal at its maximum level on the cable, and with the dipulse signal level attenuated 4dB from the nominal level at the maximum cable length and with zero cable length and no attenuator.

3.7.5.3 Bipolar. (See 4.7.5.3) The bipolar module receiver shall recover the bipolar signal from the cable at 4915.2 kb/s ± 220 b/s with no errors when the signal is transmitted over 1/4 and 1/2 mile of cable from a TD-976. The error rate shall not be greater than that specified with orderwire ring signal at its maximum level on the cable with the bipolar signal level attenuated, 4dB from the nominal level at the maximum cable length and with zero cable length and no attenuator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3.7.6 Prime power interface, (See 4.7.6) The ac primary power source shall be 115V +10% at frequencies of 50 Hz, 60 Hz, and 400 Hz ±5% single phase. The power input to the unit shall not exceed 90 watts. The ac primary power shall conform to MIL-STD-1332 for Type 1, Class 1 and 2, Model I, II, and III generator sets.

3.7.7 BITE and alarms. (See 4.7.7) The built in test equipment (BITE) shall monitor signals in and out of the unit and internal circuits and power supply voltages. The BITE shall operate without interrupting traffic or interfering with normal system operation. Removal of the BITE card shall not affect operation of the unit. Failure alarms shall be provided for loss of frame sync, channel failure, common power supply failure, and interunit signal malfunctions. Alarm circuits shall be operational within 1 second of application of power to the unit. Alarm conditions are removed automatically when proper operation is restored. Means shall be provided in the unit to check the operation of the alarms without interrupting the equipment operation.

3.7.7.1 Alarms. (See 4.7.7.1) Failures detected by the BITE circuitry shall cause relay contact closures accessible externally at a connector on the rear panel. The following outputs shall be available.

a. S alarm. The S alarm shall indicate any failures detected including loss of input power. This alarm also provides a visible indication of the front panel except for loss of input power.

b. SA alarm. The SA alarm shall indicate that the malfunction is within the unit or the unit has lost input power. Loss of input cable signal inhibits the SA alarm. If the power switch is set to OFF, there shall be no SA alarm.

c. LS alarm. The LS alarm shall indicate loss of signal coming in on any line input. If the unit loses power, the LS alarm assumes the normal state. This alarm inhibits the SA alarm on the failed group.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

3.7.8 Operational check. (See 4.7.8)

3.7.8.1 Group data error check. (See 4.7.8) There shall be no errors for an observation period of at least 10 seconds.

3.7.8.2 Digital orderwire error check. (See 4.7.8) There shall be no errors for an observation period of at least 30 seconds at 16 kb/s and at least 30 seconds at 2 kb/s.

3.7.8.3 Dipulse and Bipolar maiantenance OW output check.(See 4.7.8) The MOW output level with -4 dBm at 1.6 kHz input shall be -7 to -2 dBm looped back through no more than 10 feet of CX-11230 cable.

3.7.9 Environmental operational test. (See 4.7.9)

3.7.9.1 Maintenance OW output check. The receive level shall be between -7 and -2 dBm with the audio oscillator set to -4dBm at 1.6 kHz and the line side signal looped back through no more than 10 feet of CX-11230 cable.

3.7.9.2 Group data error check with ring tone. The dipulse and bipolar error rate measured over a 10 second period shall be zero with a -4dBm, +1.6 kHz ring tone present.

3.7.9.3 Diphase group error check. (See 4.7.9.3) There shall be no errors for an observation period of at least 10 seconds.

3.7.9.4 Digital voice orderwire check. The DVOW error rate measured over a 30 second period shall be zero.

3.7.9.5 Data orderwire error check. The DOW rate measured over a 30 second period shall be zero.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.7.9.6 Alarm monitor check. There shall be no alarms generated.

3.8 Service conditions. (See 4.8) The operational test of 3.7.9 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +63°C (+145°F), and shall withstand exposure (non-operating) to ambient air temperatures as high as +71°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient air temperatures as low as -31.7°C (-25°F), and shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall withstand, in both operating and non-operating conditions, the effects of dust particles that may accumulate within the shelter assembly or vehicle.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of a salt fog atmosphere.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Vibration and shock. (See 4.8.7). The equipment shall withstand vibration and shock induced during vehicular transport as part of a mobile assemblage, over all types of roads and cross-country terrain, and shocks encountered during servicing.

3.9 Electromagnetic interference (EMI). (See 4.9) The unit shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4.

CE02	CS02	RE02	RS02
CE03	CS06	RE02.1	RS03
CE04			
CE05			

NOTES:

a. RE02.1 is applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 is applicable over the frequency range 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to 60 Hz and 400 Hz power lines.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The unit shall have the following specified Mean-Time-Between-Failures (MTBF),θo, rated when operating in the normal modes of operation.

a. Four (4) diphase modules, plus case, power supply and BITE, 8400 hours.

b. Four (4) dipulse modules, plus case, power supply and BITE, 8400 hours.

c. Four (4) bipolar modules, plus case, power supply and BITE, 8400 hours.

d. Any combination of four (4) of the above three types of modules, plus case, power supply and BITE, 8400 hours.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.

3.12 Unit subassembly electrical tests. (See 4.12) The unit subassemblies listed below shall meet the requirements of their respective test requirement specification (TRS) drawings, as referenced in the following subassembly drawings:

CCA	TRS
BITE 1	*SM-D-875948
Conditioned Diphase A	SM-D-875969
Dipulse	SM-D-875971
Bipolar	SM-D-875972
Cable Orderwire	SM-D-875991
Conditioned Diphase B	SM-D-875992
Multi Vdc	SM-D-884567
AC Input	SM-D-884579
5 Vdc	SM-D-884593

*This drawing contains automatic test equipment (ATE) acceptance procedures and programs.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

3.13 Operational tests. Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.8 after all other inspections and tests are completed to determine that the unit is operating prior to packaging. The operational test of Group A satisfies this requirement for those units for which that is the last inspection or test.

3.14 Systems safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

3.16 Nuclear survivability. (See 4.16) The unit shall be designed to satisfy the nuclear survivability requirements stated in Section 3 of CR-CX-0173-001 with the exception of paragraphs 3.3.1.1 (BLAST) and 3.3.1.2 (THERMAL).

3.17 TEMPEST. The unit shall be designed to comply with the TEMPEST requirements specified in CR-CX-0174-001.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirments as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article inspection, (See 4.3) does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging.

(2) Packaging inspection. (See 4.14)

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 20 first article units of the complete Digital Data Modem MD-1026(P)/G. The first article quantity shall be as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspections shall consist of inspections specified in Table I and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo in-process Group A inspection prior to burn-in.

4.3.3 First article inspection data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

TABLE I. First article inspections.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
1. Inspections by Subsidiary Documents	N/A	4.4	Inspections to be performed on all units.												
2. Burn-in	3.6	4.6	Inspections to be performed on all units.												
3. Group A Inspection	(See Table II)		Inspections to be performed on all units.												
4. Group B Inspection	(See Table III)		Inspections to be performed on all units.												
5. Group C Inspection															
Subgroup 1															
High temperature	3.8.1.1	4.8.1.1	1 *												
Low Temperature	3.8.1.2	4.8.12	2 *												
Subgroup 2															
EMI	3.9	4.9			1										
TEMPEST	3.17	4.17			2										
Vibration/Shock	3.8.7	4.8.7				1 *									
Subgroup 3															
Humidity (Note C)	3.8.2	4.8.2						1			*				
Subgroup 4															
Altitude	3.8.3	4.8.3							1		*				
Dust (Note C)	3.8.4	4.8.4							2		*				
Salt Fog (Note C)	3.8.5	4.8.5							3						
Fungus (Note C)	3.8.6	4.8.6								1					
Subgroup 5															
Line to Ground Voltage	3.7.1.2	4.7.1.2										1	**		
Clock/Data Phasing	3.7.1.6	4.7.1.6										2			
Offset	3.7.1.7	4.7.1.7										3	**		
Signal Balance	3.7.1.3	4.7.1.3										4			
Slew Rate	3.7.1.8	4.7.1.8										5			

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspections. (Continued)

Inspection	Rqmt Para	Test Para	Unit 1 2	3 4 5	6 7 8 9	10 11	12-19	20+
Input Load Impedance	3.7.1.9	4.7.1.9				6		
Clockwidth Tolerance	3.7.1.5	4.7.1.5				7		
Output Rise and Fall Times	3.7.2.2	4.7.2.2				8		
Output Impedance	3.7.2.3	4.7.2.3				9		
Diphase Signals	3.7.2.4.1	4.7.2.4.1				10		
Bipolar Signals	3.7.2.4.2	4.7.2.4.2				11		
Unloaded Output	3.7.1.10	4.7.1.10				12		
Interchangeability	3.11	4.11				13		
Frequency Response	3.7.3.4	4.7.4.4				14 **		
Prime Power	3.7.6	4.7.6				15		
Asymmetry	3.7.2.5	4.7.2.5				16		
Diphase MOW Input Impedance	3.7.3.1.3	4.7.3.1.3				17		
Diphase MOW Signal to Distortion Ratio	3.7.3.1.4	4.7.3.1.4				18 **		
Diphase MOW Output Impedance	3.7.3.1.5	4.7.3.1.5				19		
Diphase and Bipolar MOW Input Impedance	3.7.3.2.3	4.7.3.2.3				20		
Dipulse and Bipolar MOW Output Impedance	3.7.3.2.4	4.7.3.2.4				21		
Full Duplex Signal Detection	3.7.5	4.7.5				22		
Power Loop	3.7.4	4.7.4				23 **		
6. Group D inspection								
Reliability	3.10	4.10					8 units	

* Indicates use of an auxiliary unit.

** One time engineering test to be covered by separate engineering report. Not part of First Article test report.

Source: https://assist.dla.mil -- Downloaded: 2026/0-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspections. (Continued)

NOTES:

a. The inspections 1 through 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by inspections 5 and 6 specify the order of inspections for the indicated units.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. The quantity of units shall be as specified in the contract or purchase order. All first article units shall be subjected to Group A and Group B inspections.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

	Where required
Printed Wiring	3.2.3
Semiconductor Devices	3.2.4
Parts, materials, and processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in Table II shall be inspected for conformance to the inspections specified in table II.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in table II. The order of inspection of the subassemblies, assemblies and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table II shall be subjected to 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table II.

b. Only for spares which are not installed into complete units, do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspection of step a shall be shall be assembled into the end equipments, which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and an end equipment 100% visual/mechanical and electrical inspection in accordance with table II in that order.

d. After completion of the above 100% inspection by the contractor discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table II. Group A electrical inspection shall be performed in the order shown in table II.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE II. Group A inspections.

Inspecton	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
VISUAL-MECHANICAL (Including Spare Assemblies)				
Multi VDC	3.15	4.13	2.5%	4.0%
AC Input	3.15	4.13	2.5%	4.0%
Conditioned Diphase A	3.15	4.13	2.5%	4.0%
Dipulse	3.15	4.15	2.5%	4.0%
Bipolar	3.15	4.15	2.5%	4.0%
BITE 1	3.15	4.15	2.5%	4.0%
Conditioned Diphase B	3.15	4.15	2.5%	4.0%
Cable OW	3.15	4.15	2.5%	4.9%
5 Vdc	3.15	4.15	2.5%	4.0%
Case and Covers	3.15	4.15	2.5%	4.0%
ELECTRICAL PERFORMANCE (Including Spare Assemblies)				
Multi Vdc	3.12	4.12	1.0%	*
AC Input	3.12	4.12	1.0%	*
Conditioned Diphase A	3.12	4.12	1.0%	*
Dipulse	3.12	4.12	1.0%	*
Bipolar	3.12	4.12	1.0%	*
BITE 1	3.12	4.12	1.0%	*
Conditioned Diphase B	3.12	4.12	1.0%	*
Cable OW	3.12	4.12	1.0%	*
5 Vdc	3.12	4.12	1.0%	*
ELECTRICAL				
BITE and Alarms	3.7.7	4.7.7	1.0%	*
Operational Tests	3.7.8	4.7.8	1.0%	*

*All electrical defects are major.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in Table III and the inspection level shall be S-4.

4.5.2.2 Order of inspecton within Group B. Group B inspection shall be performed in the order listed in table III.

TABLE III. Group B inspections.

Inspection	Reqt. Para	Insp. Para	AQL
Inspection			
Line-to-Line Voltage Levels	3.7.1.1	4.7.1.1	4.0%
Receiver Sensitivity	3.7.1.4	4.7.1.4	4.0%
Output Voltage Levels	3.7.2.1	4.7.2.1	4.0%
Diphase MOW Cable Side Output	3.7.3.1.1	4.7.3.1.1	4.0%
Diphase MOW Equipment Side Output	3.7.3.1.2	4.7.3.1.2	4.0%
Diphase and Bipolar MOW Cable Side Output	3.7.3.2.1	4.7.3.2.1	4.0%
Diphase and Bipolar MOW Equipment Side Output	3.7.3.2.2	4.7.3.2.2	4.0%
Combined Digital OW Cable-side Output	3.7.3.3	4.7.3.3	4.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units from lots that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eight months. The unit selected every fourth month shall be subjected to vibration and shock testing in that order. The unit selected every eight month shall be subjected to EMI and TEMPEST testing.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every twelfth month. One unit shall be subjected to altitude, dust and salt fog testing, and one unit shall be subjected to fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in Table IV.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed for each subgroup in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE IV. GROUP C inspections.

Inspection	Reqt Para	Insp Para
Subgroup 1		
High Temperature	3.8.1.1	4.8.1.1
Low Temperature	3.8.1.2	4.8.1.2
Subgroup 2		
EMI	3.9	4.9
TEMPEST	3.17	4.17
Vibration	3.8.7.1	4.8.7.1
Shock	3.8.7.2	4.8.7.2
Subgroup 3		
Humidity	3.8.2	4.8.2 (Note 1)
Subgroup 4		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4 (Note 1)
Salt Fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)
Subgroup 5		
Clock/ Data Phasing	3.7.1.6	4.7.1.6
Signal Balance	3.7.1.3	4.7.1.3
Slew Rate	3.7.1.8	4.7.1.8
Input Load Impedance	3.7.1.9	4.7.1.9
Clockwidth Tolerance	3.7.1.5	4.7.1.5
Unloaded Output	3.7.1.10	4.7.1.10
Output Rise and Fall times	3.7.2.2	4.7.2.2
Output Impedance	3.7.2.3	4.7.2.3
Diphase Signals	3.7.2.4.1	4.7.2.4.1
Bipolar Signals	3.7.2.4.2	4.7.2.4.2
Asymmetry	3.7.2.5	4.7.2.5
Diphase MOW Input Impedance	3.7.3.1.3	4.7.3.1.3
Diphase MOW Output Impedance	3.7.3.1.5	4.7.3.1.5
Dipulse and Bipolar Input Impedance	3.7.3.2.3	4.7.3.2.3

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE IV. Group C inspections. (Cont'd)

Inspection	Reqt Para	Insp Para
Dipulse and Bipolar MOW Output Impedance	3.7.3.2.4	4.7.3.2.4
Full Duplex Signal Detection	3.7.5	4.7.5
Interchangeability	3.11	4.11
Prime Power	3.7.6	4.7.6

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is reworked to specification and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table V and shall be performed on units from lots which have been subjected to and have met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE V. Group D inspection.

Inspection	Reqt Para	Insp Para
Reliability	3.10	4.10

4.5.5.1 Sampling for Group D inspections. Eight units shall be selected at random from the first month's production and subjected to the test of 4.10. For subsequent Group D inspections, eight units shall be selected at random from every fourth month's production.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is reworked to specification and the sample units are resubjected to and pass Group A inspection only.

4.6 Electrical preconditioning (burn-in). (See 3.6) Connect the equipment as shown in figure 1. Connections may vary depending on the card mix in the units. The test conditions of 4.10.3 apply except that duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. The requirements of 3.6 shall be met.

a. Set the Group Simulator to the appropriate group rates, with no frame format and random data.

b. Determine the type of CCA module in each group of the UUT. Set the Group Simulator and UUT switches as follows:

CCA TYPE	FUNCTION	UUT SWITCHES	GROUP SIMULATOR
DIPHASE A DIPHASE B	GROUP RATE ORDERWIRE	12(2304 kbps) SYS	12(2304 kbps)
DIPULSE	GROUP RATE TRF TRANS MI REC MI	12(2304 kbps) NORM 1/2 (internal) 1/2 (internal)	12(2340 kbps)
BIPOLAR	GROUP RATE TRF TRANS MI REC MI	15 (4915.2 kbps) NORM 1/2 (internal) 1/2 (internal)	15(4915.2 kbps)

c.Turn power on to all UUTs. Run the units for 48 hours, using the BITE to indicate any failures. Means shall be provided latch the S, SA nd LS alarms.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

d. After completion of burn-in, perform the operational tests of 4.7.8 on each UUT.

4.7 Electrical. Unless otherwise specified all tests shall be performed at 115V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipments listed in table VI or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662A. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see table VII) shall be provided by the contractor subject to approval by the Government. The Government shall be responsible for supplying the Government Furnished Equipment (GFE) listed in table VIII as required for tests. table IX identifies the external interface connections of the unit with the test figures. Unless otherwise noted, all resistors shown in the figures are 0.5 watt.

TABLE VI. Standard test equipment.

Description	Mfg/Model No.
Electronic Counter (3 Required)	H.P. 5248L
Volt-Ohm Meter (3 Required)	Simpson 260
DC Milliammeter	H.P.412A
DC Voltmeter	H.P.412A
Transmission Measuring Set (2 Required)	H.P.3555B
Signal Generator	H.P. 606B
RF Voltmeter	H.P.3406A
RMS Voltmeter (3 Required)	H.P.3404A
RMS Voltmeter	H.P. 3403A
Audio Oscillator	H.P.200CD
Audio Oscillator	H.P 201C
Wattmeter	Weston 432
Attenuator	H.P. 4436A
AC Voltmeter	H.P. 403B
Attenuator	Kay 3-30
Oscilloscope (2 Required)	Tektronix 454
Oscilloscope	Tektronix 7904
Distortion Analyzer	H.P.333A

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE VII. Special test equipment.

Equipment Name	Drawing No.	Functional Description
Group Simulator	A3051667	The Group Simulator simulates the group data, orderwire and telemetry signals. It generates either a long or short pseudo-random sequence. The three signals can be gererated simultaneously or in any combination.
Timing Source	A3051666	The Timing Source provides four independent group outputs, switch-selectable to positions between 72 and 4608 kb/s. Also provided are four independent Station Clock/Telemetry Clock outputs, switch-selectable to 2, 16, or 32 kHz.
Timing Source	(Alternate)	An alternate Timing Source is shown in figure 4 that may be used in lieu of the above.
DVOW/DOW Simulator	A3051669	The DVOW/DOW Simulator provides a pseudo-random data generator and correlator at 2, 16, and 32 kbps with a balanced signal interface A TTL error output signal is provided for counting errors externally.
Balanced Side and Diphase Side Asymmetry Generator	A3051670	Provides a balanced NRZ signal and a conditioned diphase signal which can be adjusted to produce ±5% pulse width asymmetry.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

TABLE VIII. Government furnished equipment

Equipment	Description
Multiplexer, TD-754/G	The TD-754 cable combiner provides for transmission of PCM traffic and order-wire over cable, and radio systems and provides power for Pulse Form Restorers TD-206/G's used in a cable system.
Multiplexer, TD-976/G	The asynchronous digital combiner TD-976, provides for multiplexing and de-multiplexing on a time division basis up to eight separate digital bit streams (PCM groups) into a 4.9152 Mb/s bit stream.
Cable Assembly, CX-11230/G	Dual coax cable in 1/4 mile lengths on reels.
Generator, Digital Data, SG-1139/G	The SG-1139 consists of a transmitter, which generates balanced NRZ, unbalanced NRZ or diphase test pattern signals at rates 128-4608 kb/s, 576 kb/s, and .6-32 kb/s, respectively, and a receiver, which accepts the test pattern, and detects and displays error count.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE IX MD-1026 interface connectors.

Connector	Function	HI	LO	GND	Signal Type
Line Signal In 1	Line Data In #1	-	-	-	
Line Signal Out 1	Line Data Out #1	-	-	-	
Line Signal In 2	Line Data In #2	-	-	-	
Line Signal Out 2	Line Data Out #2	-	-	-	
Line Signal In 3	Line Data In #3	-	-	-	
Line Signal Out 3	Line Data Out #3	-	-	-	
Line Signal In 4	Line Data In #4	-	-	-	
Line Signal Out 4	Line Data Out #4	-	-	-	
	Group Data In	B	A	M	NRZ
Group 1	Group Tim In	D	C	N	SQ WV
Group 2					
Group 3	Group Data Out	F	E	P	NRZ
Group 4					
	Group Tim Out	H	G	R	SQ WV
	Black Sta Clk Out	K	J	S	SQ WV
	GR 1 DVOW Data In	B	A		NRZ
	Xmt Tim Out***	D	C		SQ WV
	Data Out	F	E		NRZ
	Rcv Tim Out	H	G		SQ WV
	GR 2 DVOW Data In	K	J		NRZ
	Xmt Tim Out ***	M	L		SQ WV
	Data Out	P	N		NRZ
	Rcv Tim Out	S	R		SQ WV
Digital Voice				j,k	
Orderwire	Gr 3 DVOW Data In	U	T		NRZ
	Xmt Tim Out ***	W	V		SQ WV
	Data Out	Y	X		NRZ
	Rcv Tim Out	a	Z		SQ WV
	Gr 4 DVOW In	c	b		NRZ
	Xmt Tim Out ***	e	d		SQ WV
	Data Out	g	f		NRZ
	Rcv Tim Out	i	h		SQ WV
Analog Voice Orderwire	Gr 1 AVOW In	B	A	Analog	
	AVOW Out	D	C		Analog
	Gr 2 AVOW In	F	E		Analog
	Avow Out	H	G	T,U	Analog

*** Fiber optics modules only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE IX MD-1026 Interface connectors (contd)

Connector	Function	Pin HI	Pin LO	Pin GND	Signal Type
ANALOG VOICE ORDERWIRE (contd)	Gr 3 AVOW In	K	J		Analog
	AVOW Out	M	L		Analog
	Gr 4 AVOW	P	N		Analog
	AVOW Out	S	R		Analog
DATA ORDERWIRE	GR 1 DOW Data In	B	A		NRZ
	Xmt Tim Out	D	C		SQ WV
	Data Out	F	E		NRZ
	Rcv Tim Out	H	G		SQ WV
	GR 2 DOW Data In	K	J		NRZ
	Xmt Tim Out	M	L		SQ WV
	Data Out	P	N		NRZ
	Rcv Tim Out	S	R		SQ WV
	Gr 3 DOW Data In	U	T		NRZ
	Xmt Tim Out	W	V		SQ WV
	Data Out	Y	X		NRZ
	Rec Tim Out	a	Z		SQ WV
	Gr 4 DOW Data In	c	b		NRZ
	Xmt Tim Out	e	d		SQ WV
	Data Out	g	f		NRZ
	Rcv Tim Out	i	h		SQ WV
STATION CLOCK and ALARM	Alarm S	B	A		Alm=Closed Contacts
	Alarm SA	D	C		Alm=Closed Contacts
	Alarm LS	F	E		Alm=Closed Contacts
	Station Clock In	K	G	H,J	SQ WV
TO ROA 1 TO ROA 2 TO ROA 3 TO ROA 4	XMTR Traffic Fault	A			±4 VDC
	Optic Fault	B			±4 VDC
	RCV Traffic Fault	C			±4 VDC
	+5	D		H,K,J	DC Power
	-5	E			DC Power
	+12	F			DC Power
	-12	G			DC Power
POWER		A	C	B	

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.1 Balanced equipment-side signal characteristics.

4.7.1.1 Line-to-line voltage levels diphase, dipulse and bipolar signals.

a. Determine the type of module* in each group of the UUT. Set the Group Simulator and UUT switches as shown in table IX-1

Table IX-1 UTT switch configuration.

CCA Type	Function	UUT Switches	Group Simulator
Diphase A Diphase B	Group Rate Orderwire	14 (4608 kbps) SYS	14 (4608 kbps)
Dipulse	Group Rate TRF Trans MI Rec MI	12 (2304 kbps) Norm 1/2 (internal) 1/2 (internal)	12 (2304 kbps)
Bipolar	Group Rate TRF Trans MI Rec MI	15 (4915.2 kbps) Norm 1/4 (internal) 1/4 (internal)	15 (4915.2 kbps)

b. For those groups or modules containing diphase CCAs connect the Group Simulator and DVOW/DOW simulators to the first diphase group as shown in figure 3.

c. For each signal shown in table X, measure each output signal with the oscilloscope connected as in figure 4(A).

d. For an all "1" pattern, and all "0" pattern and a random pattern, verify that the line-to-line voltage level requirements of 3.7.1.1 are met.

e. Repeat b through d for each remaining Diphase module.

* A Diphase module or group contains one Diphase A CCA and one Diphase B CCA.
A Dipulse module or group contains one Dipulse CCA and one Cable OW CCA.
A Bipolar module or group contains one Bipolar CCA and one Cable OW CCA.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE X Diphase Group/OW Signals

Group data out	14(4608 kbps)
Group timing out	14(4608 kbps)
Black station clock out	14(4608 kbps)
DVOW data out	16 kbps
DVOW timing out	16 kbps
DOW data out	2 kbps
DOW Xmit timing out	2 kbps
DOW received timing out	2 kbps

f. For those groups containing Dipulse and Bipolar modules, connect only the Group Simulator to the first Dipulse or Bipolar groups as shown in figure 3.

g. Set the Group Simulator to its appropriate bit rate (see step a) and perform steps c and d for the output signals shown in table XI.

h. Repeat steps f and g for the remaining dipulse and Bipolar groups.

Table XI Dipulse/Bipolar group output signals

	Dipulse	Bipolar
Group data out	12(2304 kbps)	15(4915.2 kbps)
Group timing out	12(2304 kbps)	15(4915.2 kbps
Black station clock out	12(2304 kbps)	15(4915.2 kbps)

4.7.1.2 Line-to-ground voltage, Diphase, Dipulse, and Bipolar. (See 3.7.1.2)

a. The equipment setup is the same as in 4.7.1.1.

b. For each of the output signals shown in table X at the rates indicated, measure the voltage levels on each HI and LOW line with respect to ground on all groups containing Diphase modules using the oscilloscope as in figure 4(B).

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

c. Verify that the voltage levels are within the required limits of 3.7.1.2.

d. For each Dipulse and Bipolar module installed measure the line-to-ground voltage as in step b for each output signal shown in table XI at the bit rates indicated.

e. Verify that the line-to-ground voltage requirements of 3.7.1.2 are met.

4.7.1.3 Signal balance, Diphase, Dipulse and Bipolar. (See 3.7.1.3)

a. The equipment setup and procedure is the same as in 4.7.1.1

b. For each Diphase module installed, measure the positive and and negative voltage levels line-to-line of each signal shown in table X with the oscilloscope CH-A and CH-B input differentially referenced to ground.

c. Compare the magnitudes of the positive and negative parts of the waveform. Determine the larger of the two values.

d. Take the magnitude of the difference between the positive and negative values and compare with 10% of the magnitude of the larger value.

e. Verify that the requirements of 3.7.1.3 are met.

f. For each Dipulse and Bipolar module installed measure the line-to-line voltage as in step b for each signal shown in table XI.

g. Using steps c and d above, verify that the signal balance requirements of 3.7.1.3 are met.

4.7.1.4 Receiver sensitivity Diphase, Dipulse and Bipolar. (See 3.7.1.4)

a. Connect the equipment as shown in figure 5 with a variable attenuator inserted in the normal signal input paths.

b. Determine the type of CCA module in each group of the UUT. Set the Group Simulator and UUT switches as shown in table IX-1.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

c. Connect the Group and DVOW/DOW Simulators to the first Diphase group in the UUT and set the group rate as shown in step b.

d. Adjust the attenuators to give a +50 mV peak signal on each signal shown below:

Group data in	14 (4608 kbps)
Group timing in	14 (4608 kbps)
DVOW data in	16 kbps
DOW data in	2 kbps
Station clock in	32 kbps

e. Verify that there are no group data errors for a period of 10 seconds and no DVOW (16 kbps) errors or DOW (2 kbps) errors for a period of 30 seconds.

f. Repeat steps c through e for each remaining Diphase module installed in the UUT.

g. Connect the Group Simulators only to the first Dipulse or Bipolar group. Set the appropriate group rate and repeat step d for each signal below.

	Dipulse	Bipolar
Group data in	12 (2304 kbps)	15 (4915.2 kbps)
Group timing in	12 (2304 kbps)	15 (4915.2 kbps)

h. Verify that there are no group data errors for a period of 10 seconds.

i. Repeat steps g and h for each remaining Dipulse and Bipolar module.

4.7.1.5 Clockwidth tolerance Diphase, Dipulse and Bipolar. (See 3.7.1.5)

a. The equipment set up is the same as in 4.7.1.1.

b. For each Diphase module installed and all signals shown in Table X, measure the time (A) between the rise and fall slopes at the zero voltage points; then measure the time (B) between two successive rise slopes at the zero voltage points as indicated in Figure 6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

c. Verify that the clock width tolerances meet the requirements of 3.7.1.5.

d. For each Dipulse and Bipolar module installed and all signals shown in Table XI, repeat steps b and c.

4.7.1.6 Clock data phasing, Diphase, Dipulse and Bipolar. (See 3.7.1.6)

a. The equipment set up is the same as in 4.7.1.1.

b. Connect the oscilloscope to the group 1 signals as shown in figure 7(A) with the two sets of channels each set up in the differential mode referenced to ground.

c. For each set of signals given below, measure the delay between the zero voltage points on both the data rise and fall time slopes to the zero voltage points of the timing rise slopes as indicated on the wave form diagram in figure 8.

	Diphase	Dipulse	Bipolar
GROUP DATA/ TIMING OUT	rate 14(4608 kbs)	rate 12(2304 kbs)	rate 15(4915.2kbs)
DVOW DATA/ TIMING OUT	16 kbs	-	-
DOW DATA/ RCV TIMING OUT	2 kbs	-	-

d. Repeat steps d and e for each remaining module in the UUT.

e. Verify that the requirements of 3.7.1.6 are met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.1.7 Offset, Diphase, Dipulse and Bipolar. (See 3.7.1.7)

a. The equipment set up is the same as in 4.7.1.1.

b. Connect the oscilloscope to the group 1 signals as shown in figure 7(B)

c. For each diphase module installed and all signals shown in table X, measure the DC offset voltage between the high and low pins.

d. Verify that the requirements of 3.7.1.7 are met.

e. For each Dipulse and Bipolar module installed and all signals shown in table XI, repeat steps d and e.

4.7.1.8 Slew rate Diphase, Dipulse and Bipolar. (See 3.7.1.8)

a. The equipment set up is the same as in 4.7.1.1.

b. Connect the oscilloscope to the group 1 signals as shown in figure 4A except that 100 ft of RG-108A/U cable shall be inserted between the oscilloscope and the UUT, and the resistor values shall be 39 ohms ±1%.

c. For each Diphase module installed and all signals shown in table X, measure the time between the +40 mV and -40 mV points around the zero voltage level of the rise and fall slopes as shown in figure 9

d. Verify that the requirements of 3.7.1.8 are met.

e. For each Dipulse and Bipolar module installed and all signals shown in table XI, repeat c and d.

4.7.1.9 Input load impedance, Diphase, Dipulse and Bipolar. (See 3.7.1.9)

a. Connect the test setup as shown in figure 10 and measure the following signals:

Group Data and Timing Input	All Types of modules
Digital Voice Orderwire Data In	Diphase module only
Data Orderwire Data In	Diphase module only

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

b. At each signal input, adjust the audio oscillator at a frequency of 1.0 kHz for an output voltage of 1.0 volt as measured by the V_1 voltmeter, then measure the V_2 voltage.

c. Calculate the input impedance of each input using the formula:

$$R = 100 \frac{V2}{V1} \text{ ohms}$$

d. Verify that the calculated input impedance meets the requirements of 3.7.1.9.

4.7.1.10 Unloaded output, 2 kb/s data and timing. (See 3.7.1.10)

a. Connect the equipment as shown in figure 11.

b. This test is only conducted on groups containing Diphase CCA's. Select the first Diphase group and connect the oscilloscope as shown without the orderwire signals terminated.

c. Set the Group Simulator and UUT switches as follows:

CCA Type	Function	Switches	Group Simulator
Diphase A	Group Rate	14(4608 kbps)	
Diphase B	Orderwire	SYS	14(4608 kbps)

d. Measure the unloaded, positive and negative voltage levels line-to-line of each output signal shown below.

DOW DATA out
DOW RCV TIM out
DOW XMT TIM out

e. Repeat step c for each remaining Diphase Group.

f. Verify that the requirements of 3.7.1.10 are met.

4.7.2 Cable-side group signal characteristics. (See 3.7.2)

4.7.2.1 Output voltage levels: Diphase, Dipulse and Bipolar signal. (see 3.7.2.1)

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

a. Connect the equipment as shown in figure 12.

b. Determine the type of module in each group of the UUT. Set the Group Simulator and UUT switches as follows:

CCA Type	Function	Switches	Group Simulator
Diphase A Diphase B	Group Rate Orderwire	14 (4608 kbps) Maint	14 (4608 kbps)
Dipulse	Group Rate TRF	12 (2304 kbps) Norm	12 (2304 kbps)
Bipolar	Group Rate TRF	15 (4915.2 kbps) Norm	15 (4915.2 kbps)

c. Diphase groups.

(1) Select the first Diphase Group of the UUT. Set up the Group Simulator to generate an all ones pattern.

(2) Measure the peak-to-peak amplitude of the output voltage at the associated line SIG OUT connector on the UUT as shown in figure 13.

(3) Set the Group Simulator to generate an all zero pattern and repeat step (2).

(4) Repeat steps (1) through (3) for each of the remaining Diphase groups in the UUT.

(5) Verify that the requirements of 3.7.2 (a) are met.

d. Dipulse groups.

(1) Select the first Dipulse group in the UUT, set up the Group Simulator to generate an all ones pattern.

(2) Measure the base-to-peak amplitude of the voltage at the LINE SIGNAL OUT connector of the UUT as shown in figure 14.

(3) Measure the pulse width as shown in figure 14.

(4) Verify that the Dipulse pulse width meets the requirements of 3.7.2.1(b).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

(5) Repeat steps 2 through 4 for any remaining Dipulse Groups in the UUT.

e. Bipolar groups.

(1) Select the first Bipolar group of the UUT, and set up the Group Simulator to generate an all ones pattern.

(2) Measure the base-to-peak amplitude of the output voltage for a positive and a negative pulse at the LINE SIGNAL OUT connector of the UUT as shown in Figure 15.

(3) Measure the pulse width of a positive and a negative pulse as shown.

(4) Verify that the requirements of 3.7.2.1 (c) are met.

(5) Repeat steps 2 through 4 for any remaining Bipolar modules in the UUT.

4.7.2.2 Output rise and fall time. (See 3.7.2.2)

a. Connect the equipment as shown in figure 12. The switch settings for the UUT are the same as in 4.7.2.1.

b. Diphase groups

(1) Select the first Diphase group of the UUT. Set the Group Simulator to generate an all zeroes pattern.

(2) Measure the rise and fall time of the diphase signal at the LINE SIGNAL OUT. Refer to Figure 13.

(3) Verify that the requirements of 3.7.2.2 (a) are met.

(4) Repeat steps (1) through (3) for each of the remaining Diphase modules in the UUT.

c. Dipulse groups.

(1) Select the first Dipulse group of the UUT. Set the Group Simulator to generate an all ones pattern.

(2) Measure the rise and fall time of the dipulse signal at the LINE SIGNAL OUT. Refer to figure 14.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

(3) Verify that the requirements of 3.7.2.2 (b) are met.

(4) Repeat steps (1) through (3) for each of the remaining Dipulse modules in the UUT.

d. Bipolar groups.

(1) Select the first Bipolar group of the UUT. Set the Group Simulator to generate an all ones pattern.

(2) Measure the rise and fall time of the bipolar signal at the LINE SIGNAL OUT. Refer to Figure 15.

(3) Verify that the requirements of 3.7.2.2 (c) are met.

(4) Repeat steps (1) through (3) for each of the remaining Bipolar modules in the UUT.

4.7.2.3 Output impedance, Diphase and Bipolar signals. (See 3.7.2.3)

a. Connect the equipment as shown in figure 16.

b. Set the UUT switch settings the same as in 4.7.2.1.

(1) Select the first Diphase or Bipolar group.

(2) Measure the signal output voltage with S1 open (V1) and with S1 closed (V2).

(3) Calculate the output impedance using the formula:

$$Z_{out} = 58\left(\frac{V1}{V2}-1\right)\text{ohms}$$

(4) Verify that the requirements of 3.7.2.3 a and are met.

(5) Repeat steps (1) through (4) for each of the remaining Diphase and Bipolar modules.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.2.4 Input impedance: Diphase and Bipolar signals. (See 3.7.2.4)

4.7.2.4.1 Diphase signals. (See 3.7.2.4.1)

a. Select the first Diphase group in the UUT and connect the test setup as shown in figure 17 to the LINE SIGNAL IN connector for that group.

b. Tune the Signal Generator in turn to 300 Hz, 1000 Hz, and 3000 kHz. At each frequency, set its output level to 100 mV as measured at V_1 by the voltmeter.

c. Measure the V_2 voltage.

d. Calculate the input impedance (Z) at each frequency level using the formula:

$$Z = 58\left(\frac{V2}{V1 - V2}\right)\text{ohms}$$

e. Verify that the calculated input impedance meets the requirements of 3.7.2.2.1.

f. Repeat steps a through e for all remaining Diphase Groups in the UUT.

4.7.2.4.2 Bipolar signals. (See 3.7.2.4.2)

a. Select the first Bipolar group in the UUT and connect the test setup as shown in figure 17 to the LINE SIGNAL IN connector for that group.

b. Tune the signal generator in turn to 400 kHz and 3 MHz. At each frequency, set its output level to 100 mV as measured at V_1 by the voltmeter.

c. Measure the V2 voltage.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

d. Calculate the input impedance (Z) at the 400 kHz and 3.0 MHz frequency level using the formula:

$$Z = 58\left(\frac{V2}{V1 - V2}\right) \text{ohms}$$

e. Verify that the calculated input impedance meets the requirements of 3.7.2.4.2.

f. Repeat steps a through e for all remaining Bipolar Groups in the UUT.

4.7.2.5 Asymmetry. (See 3.7.2.5)

4.7.2.5.1 Transmit Asymmetry. (See 3.7.2.5.1)

a. Connect the equipment as shown in figure 18, select the first Diphase Group and set the Group Rate to 4608 kb/s, (rate 14).

b. Set the Diphase B CCA OW switch to MAINT.

c. Set the group rate on the balanced asymmetry generator to 4608 kb/s.

d. Set the asymmetry out of the generator to +5%.

e. Measure the asymmetry out of the UUT.

f. Set the asymmetry output of the generator to -5%.

g. Measure the Asymmetry out of the UUT.

h. Set the group rate on the balanced asymmetry generator to 576 kb/s and the Group Rate of the Diphase A CCA to position 7 (576 kb/s).

i. Repeat Steps d through g.

j. Repeat Steps b through g for each remaining Diphase group in the UUT. Verify that the requirements of 3.7.2.5.1 are met.

Note: Positive number indicates that the middle diphase transition is early.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.2.5.2 Receive Asymmetry. (See 3.7.2.5.2)

a. Connect the equipment as shown in figure 19. Select the first Diphase group in the UUT.

b. Set Diphase B CCA OW switch to SYS.

c. For each of the SYS rates and cable length combinations shown in figure 19, perform steps d through h.

d. Set the Asymmetry to +5%.

e. Monitor the group errors for 5 minutes.

f. Verify that the requirements of 3.7.2.5.2 are met.

g. Set the Asymmetry to -5%.

h. Repeat Steps e and f.

i. Repeat steps b through h for each remaining Diphase group in the UUT.

4.7.3 Orderwire signal characteristics. (See 3.7.3)

4.7.3.1 Diphase maintenance orderwire. (See 3.7.3.1)

4.7.3.1.1 Cable-side output. (See 3.7.3.1.1)

a. Connect the equipment as shown in figure 20 to the first Diphase group in the UUT.

b. Set all Diphase A CCA Group Rate switches to OFF (Rate 0) and all Diphase B CCA OW switches to MAINT.

c. Set the oscillator frequency to 1 kHz, its output level to 977 mV and connect it to the ANALOG ORDERWIRE connector in the UUT.

d. Measure the loaded output voltage at the LINE SIGNAL OUT connector on the UUT into 58 ohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

e. Calculate the output power using the formula:

$$dBm = 20 \log V + 12.37 \text{ dBm}$$

when V is in volts rms.

f. Verify that the requirements of 3.7.3.1.1 are met.

g. Repeat steps a through f for any remaining Diphase modules.

4.7.3.1.2 Equipment-side output. (See 3.7.3.1.2)

a. Connect the equipment as shown in figure 21 and select the first Diphase group.

b. Connect the Signal Generator to the LINE SIG IN connector and adjust the generator frequency to 1 kHz and its level to 57 mV (-18.5 dBm at LINE SIG IN connector).

c. Measure the loaded output voltage at the associated ANALOG ORDERWIRE OUT connector.

d. Calculate the output power using the formula:

$$dBm = 20 \log V + 2.22 \text{ dBm}$$

e. Repeat Steps b and c with the oscillator level set to 1.21V (+8 dBm at LINE SIG IN connector).

f. Verify the requirements of 3.7.3.1.2 are met.

g. Repeat steps b through f for all remaining Diphase groups in the UUT.

4.7.3.1.3 Equipment side input impedance. (See 3.7.3.1.3)

a. Connect the equipment as shown in figure 10 with the measurement hook-up connected to the analog voice orderwire connector on the UUT and the resistor value changed to 600 ohms ±1%.

b. Set all Diphase A CCA group rate switches to OFF (rate 0) and all Diphase B CCA OW switches to°MAINT.

c. Adjust the oscillator frequency to 300 hz and set level V1 to 10V.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

d. Measure the V2 voltage.

e. Calculate the input impedance using the formula:

$$Z = 600\left(\frac{V2}{V1}\right)\text{ohms}$$

f. Verify that the requirements of 3.7.3.1.3 are met.

g. Repeat c through f, for all remaining Diphase groups.

4.7.3.1.4 Signal to distortion ratio. (See 3.7.3.1.4)

a. Connect the equipment as shown in figure 22.

b. Set all Diphase A CCA Group Rates to 14 (4608 kb/s) and all Diphase B CCA OW to MAINT.

c. Connect the Audio Oscillator to the first AVOW IN line and set the oscillator to 1 kHz at -4dBm.

d. Set the distortion analyzer meter to indicate 0 dBm.

e. Measure the distortion at AVOW OUT with the analyzer tuned to 1 kHz.

f. Verify that the requirements of 3.7.3.1.4 are met.

g. Repeat c through f for all remaining Diphase groups in the UUT.

4.7.3.1.5 Equipment side output impedance. (See 3.7.3.1.5)

a. Connect the equipment as shown in figure 23(A).

b. Set all Diphase A CCA Group Rate switches to OFF (Rate 0) and all Diphase B CCA OW switches to MAINT.

c. Connect the oscillator to the AVOW In of the first Diphase group and adjust the oscillator frequency to 300 Hz at -4 dBm.

d. Measure the AVOW OUT output open circuit voltage V_1 at the Analog Voice Orderwire connector on the UUT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

e. Connect the equipment as shown in figure 23(B).

f. With the oscillator as in step c measure the AVOW Out output voltage V_2 at the Analog Voice Orderwire connector in the UUT.

g. Calculate the impedance using the formula:

$$Z = 600\left(\frac{V1 - V2}{V2}\right)\text{ohms}$$

h. Repeat steps c through g for 1 kHz and 3400 Hz.

i. Verify that the requirements of 3.7.3.1.5 are met.

j. Repeat steps c through h for all remaining Diphase groups in the UUT.

4.7.3.2 Dipulse and bipolar maintenance orderwire. (See 3.7.3.2)

4.7.3.2.1 Cable-side output. (See 3.7.3.2.1)

a. Connect the equipment as shown in figure 24 to the first Dipulse or Bipolar group in the UUT.

b. Set all Dipulse and Diphase CCA Group Rate switches to OFF (Rate 0).

c. Set the oscillator frequency to 1 kHz, its output level to 977 mV (-4dBm at ANALOG ORDERWIRE connector)and connect it to the AVOW IN signal on the UUT.

d. Measure the loaded output voltage at the LINE SIGNAL OUT connector on the UUT into 880 ohms.

e. Verify that the requirements of 3.7.3.2.1 are met.

f. Repeat steps a through e for all remaining Dipulse and Bipolar groups in the UUT.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.7.3.2.2 Equipment-side output (See 3.7.3.2.2)

a. Connect the equipment as shown in figure 25. Select the first Dipulse or Bipolar group.

b. With the Audio Oscillator connected to the LINE SIGNAL IN connector for the first module, adjust the audio oscillator frequency to 1 kHz and its output level, V_1 to 0.9 volt as measured at the V_1 voltmeter.

c. Measure the loaded voltage, V_2, at the output of the Analog Orderwire Connector on the UUT.

d. Increase the oscillator output, V_1, level to 9.0 volts and measure the loaded voltage, V_2, at the output of the ANALOG ORDERWIRE connector on the UUT.

e. Calculate the power using the formula:

$$dBm = 20 \log V + 2.22 \text{ dBm}$$

where V is in volts rms.

f. Verify that the requirements of 3.7.3.2.2 are met.

g. Repeat steps b through f for all remaining Bipolar and Dipulse groups in the UUT.

4.7.3.2.3 Cable-side input impedance. (See 3.7.3.2.3)

a. Connect the equipment as shown in figure 26 and select the first Dipulse or Bipolar group.

b. Set the oscillator output to 500 mV rms as measured at Vl and its frequency to 300 Hz.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

c. Measure the V1 and V2 voltages at the LINE SIGNAL in connector on the UUT.

d. Set the oscillator output to 500 mV rms as measured at V_1 and its freqency to 1700 Hz.

e. Repeat step c.

f. Calculate the input impedance using the formula:

$$Z = 880\left(\frac{V2}{V1}\right) \text{ ohms}$$

g. Verify that the requirements of 3.7.3.2.3 are met.

h. Repeat steps a through g for each of the remaining Dipulse and Bipolar groups.

4.7.3.2.4 Cable-side output impedance. (See 3.7.3.2.4)

a. Connect the equipment as shown in figure 27. Select the first Dipulse or Bipolar Group. Set traffic rate to 0.

b. Set the Audio Oscillator frequency to 300 Hz and its output level to -4 dBm and connect it to the ANALOG ORDERWIRE connector on the UUT.

c. Measure the amplitude of the loaded output voltage, V_1, with S1 closed and the open circuit output voltage, V2, with S1 open at the LINE SIGNAL OUT connector.

d. Calculate the output impedance, Z, using the formula:

$$Z = 880\left(\frac{V2 - V1}{V1}\right) \text{ohms}$$

e. Verify that the requirements 3.7.3.2.4 are met.

f. Repeat steps a through e all Dipulse and Bipolar modules installed in the UUT.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.7.3.3 Combined digital orderwire cable-side output. (See 3.7.3.3).

a. Connect the equipment as shown in figure 28. Select the first Diphase group. Set the Diphase A CCA Group Rate switch to OFF (Rate 0) and the Diphase B CCA OW switch to SYS.

b. Measure the positive and negative peak amplitudes of the loaded output voltage V1, at the LINE SIGNAL OUT connector of the UUT for the first Diphase module installed.

c. Verify that the requirements of 3.7.3.3 are met.

d. Repeat steps b and c for all remaining Diphase modules installed.

4.7.3.4 Frequency response. (See 3.7.3.4)

a. Connect the equipment as shown in figure 29. Select the first Diphase group in the UUT.

b. Set all Group Traffic Rate switches on the Diphase A CCA's to OFF and the Diphase B CCA's OW switch to MAINT.

c. With the oscillator connected to the group AVOW Input, set the oscillator frequency to 1 kHz, its output level to 45 mV rms.

d. Maintain the oscillator voltage constant and set the oscillator to each of the frequencies listed below. At each frequency measure the AVOW output level.

FREQUENCY
300
400
3000
3400

e. Verify that the requirements of 3.7.3.4. are met.

f. Connect the oscillator to the first Dipulse group in the UUT. Set the oscillator frequency to 1 kHz and the output level to 977 mV (equivalent to -4 dBm into the unit).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

g. Maintain the oscillator voltage constant and set the oscillator to each of the frequencies listed below. At each frequency, measure the AVOW output level.

FREQUENCY
300
1700

h. Verify that the requirements of 3.7.3.4 are met.

i. Repeat steps f through h for each Bipolar module installed in the UUT.

4.7.4 Power loop and current protection. (See 3.7.4)

4.7.4.1 Power loop: Dipulse and Bipolar Groups.

a. Connect the equipment as shown in figure 30 (A). Ensure that the positive terminal of the power supply is connected to the LINE SIGNAL IN connector on the UUT for the first Dipulse or Bipolar module installed.

b. Adjust the power supply output until the milliammeter shows a 45 mA current flow.

c. Measure the voltage drop on the voltmeter.

d. Verify that the requirments of 3.7.4 are met.

e. Repeat steps b through d for each Dipulse and Bipolar module installed.

4.7.4.2 Current protection.

a. Set each group to its maximum bit rate (Dipulse 2304 kbps, Bipolar 4915.2 kbps, Diphase A 4608 kbps) and Diphase B CCA OW switch to SYS.

b. Connect the MD-1024/G LINE SIGNAL OUTPUT to each UUT cable side input and output as shown in figure 30 (B).

c. Turn on the MD-1024/G cable current for a minimum of 1 minute.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

d. Set Diphase B CCA OW switch to MAINT and repeat steps b and c.

e. Perform operation check of 4.7.8.

4.7.5 Full duplex signal detection, dynamic range, and timing source tests. (See 3.7.5)

4.7.5.1 Diphase tests. (See 3.7.5.1)

a. Connect the equipment as shown in figure 31 and select a Diphase Group.

b. Adjust the Audio Oscillator output level to -3 dBm and its frequency to 1600 Hz.

c. With the CX-11230/G cable lengths and attenuation, Orderwire switch positions, Group Simulator Rates, and Diphase A CCA Group Rate switch settings given below, monitor the error count for 10 seconds.

d. Verify that the requirements of 3.7.5.1 are met.

DIPHASE COMBINATIONS

Cable Length, km	UUT and Grp Simulator Bit Rate, kb/s	UUT Orderwire Switch Position
3.2 + 4 dB	144 (3)	MAINT
3.2 + 4 dB	72 (1)	MAINT
0 + 0 dB	144 (3)	MAINT
0 + 0 dB	72 (1)	MAINT
0.8 + 4 dB	4608 (14)	SYS
1.6 + 4 dB	2304 (12)	SYS
3.2 + 4 dB	576 (7)	SYS
3.2 + 4 dB	256 (4)	SYS
0 + 0 dB	4608 (14)	SYS
0 + 0 dB	256 (4)	SYS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.5.2 Dipulse Tests. (See 3.7.5.2)

a. Connect the equipment as shown in figure 32 and select one Dipulse group.

b. Adjust the audio oscillator output level to -4 dBm and its frequency to 1600 Hz.

c. With the CX-11230/G cable lengths and attenuation, the Group Simulator rate, Dipulse Rate and Miles switch setting given below, monitor the error count for 10 seconds.

d. Verify that the requirements of 3.7.5.2 are met.

DIPULSE COMBINATIONS

Cable Length, km	UUT and Grp Simulator Bit Rate, kb/s	UUT Mile Switch Position XMIT	RELV
1.6 + 4 dB	2304 (12)	4	4
1.6 + 4 dB	288 (5)	4	4
1.2	2304 (12)	3	1
0.8	2304 (12)	2	4
0.4	2304 (12)	1	3
0	2304 (12)	2	2

4.7.5.3 Bipolar tests.

a. Connect the equipment as shown in figure 33 and select one bipolar group and set the SG-1139 to ATACS 576 kb/s and loop timing.

b. Adjust the Audio Oscillator #1 output level to -4 dBm at 1000 Hz and adjust Audio Oscillator #2 to 224 $\pm$ 2 mV at 1000 Hz.

c. With the CX-11230/G cable lengths and attenuation, traffic rate, and the Miles switch settings given below monitor the error count for 10 seconds.

d. Verify that the requirements of 3.7.5.3 are met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

BIPOLAR COMBINATIONS

	UUT and GRP SIM	UUT Sw Pos		
Cable Length, km	Bit Rate, kb/s	XMT	RCV	TRF
0.8 + 4 dB	4915.2 (15)	1/2	1/2	NORM
0.4	4915.2 (15)	1/4	1/4	NORM
0	4915.2 (15)	1/4	1/4	NORM

4.7.6 Prime power variation. (See 3.7.6)

4.7.6.1 Diphase CCAs in all slots. For this test it will be necessary to configure the UUT with Diphase CCAs in all groups.

a. Connect the equipment as shown in figure 34. For each Diphase A CCA set the group rate to 14 (4608 kbs) and for each Diphase B CCA set Orderwire switch to SYST. Set Group Simulator Rate to 14 (4608 kbs).

b. For each of the combinations of voltage and frequency given below,

(1) Measure the input power.

(2) Monitor the Group ERRORS over a 10 second period.

(3) Monitor the DVOW and DOW ERRORS over a 30 second period.

c. Verify that the requirements are met.

Voltage	115 VAC	103.5 VAC		126.5 VAC	
Frequency	60 Hz	47.5Hz	420Hz	47.5Hz	420Hz

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.6.2 Dipulse CCAs in all slots.

a. For each Dipulse CCA set Group Rate to 12 (2304 kbs), TRF to NORM, Trans MI to 2 (1/2), and RCV MI to 4 (1/2). Set the Group Simulator Rate to 12(2304 kbs.)

b. For each of the combinations of voltage and frequency give below

(1) Measure the input power.

(2) Monitor the Group errors over a 10 second period.

c. Verify that the requirements of 3.7.6 are met.

Dipulse tests.

Test Conditions

Voltage	115V	103.5V	126.5V	126.5V	103.5V
Frequency	60 Hz	47.5 Hz	47.5 Hz	420 Hz	420 Hz

4.7.6.3 Bipolar CCAs in all slots.

a. For each Bipolar CCA set Group Rate to 15 (4915.2 kbs), TRF to NORM, Trans MI to 1/2, and RCV MI to 1/2. Set the Group Simulator group rate 15 (4915.2 kbs).

b. For each of the combinations of voltage and frequency given below:

(1) Measure the input power.

(2) Monitor Group errors over a 10 second period.

c. Verify that the requirements of 3.7.6 are met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

Bipolar tests.

Test condition

Voltage	115V	103.5V	126.5V	126.5V	103.5V
Frequency	60 Hz	47.5 Hz	47.5Hz	420 Hz	420 Hz

4.7.7 BITE and alarms tests. (See 3.7.7) The Group Modem accepts any combination of Diphase, Dipulse or Bipolar modules in any group position of the Group Modem Case. For test purposes configure the UUT as follows:

a. Put ANY or ALL Diphase modules in the first Group Locations.

b. Put ANY or ALL Dipulse modules in the next Group Locations.

c. Put ANY or ALL Bipolar modules in the last Group Locations.

4.7.7.1 Diphase.

a. Connect the equipment as shown in figure 35 for the Diphase module.

b. Set the Group Simulator and UUT Diphase A CCA Group Rate Select Switch to 14 (4608 kb/s). Set all other groups to OFF (Rate 0).

c. Verify that the following BITE and Alarm conditions are met, (Open Circuit is the Normal state. Closed indicates alarm).

Requirement

S Open

SA Open

LS Open

Summmary fault lamp must be OFF

All CCA LEDs must be OFF

ALL Power supply fault lights must be OFF

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

d. Press the LAMP TEST switch on the UUT.

e. Verify that the BITE requirements listed below are met:

Requirement
S Closed
LS Closed
SA Open
Summary fault, Power Supply lamp, and loss at Input lamp must light
All CCA Fault lamps light

f. Disconnect the loop back cable from the Diphase module Line Signal Out connector on the UUT.

g. Verify that the BITE requirements below are met:

Requirement
S Closed
LS Closed
SA Open
Summary fault lamp must light
Diphase A CCA fault must light

h. Reconnect the Loop Back to the LINE SIGNAL OUT connector on the UUT.

i. Remove BITE card and observe that there are no errors.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.7.2 Dipulse.

a. Connect the equipment as shown in figure 35 for the Dipulse module.

b. Set the Group Simulator and UUT Dipulse CCA Rate Select switch to 12(2304 kb/s) the internal switches S3 and S4 to 1/2 mile. Set all other groups to OFF (Rate 0).

c. Repeat 4.7.7.1 steps c through i.

4.7.7.3 Bipolar.

a. Connect the equipment as shown in figure 35 for the Bipolar modules.

b. Set the Group Simulator and UUT Bipolar CCA rate switch to 15 (4915.2 kb/s), set XMT and RCV miles switches to 1/4 miles. Set all other groups to OFF (Rate 0).

c. Repeat 4.7.7.1 steps c through i.

4.7.7.4 Loss of power.

a. Disconnect power to the UUT (leave the power switch in the ON position).

b. Verify that the BITE requirements listed below are met:

Requirement
S Closed
LS Open
SA Closed

c. Turn off the unit power switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

d. Verify that the BITE requirements listed below are met:

	Requirement
S	Closed
LS	Open
SA	Open

4.7.8 Operational check. (See 3.7.8)

a. Connect the equipent as shown in figure 36.

b. Determine the type of module in each group of the UUT. Set the Group Simulator and UUT switches as follows:

CCA Type	Function	UUT Switches	Group Simulator
DIPHASE A DIPHASE B	GROUP RATE ORDERWIRE	14 (4608kbps) SYS	14
DIPULSE	GROUP RATE TRF TRANS MI REC MI	12 (2304 kbps) NORM 1/2 (internal) 1/2 (internal)	12
BIPOLAR	GROUP RATE TRF TRANS MI REC MI	15 (4915.2 kbps) NORM 1/2 (internal) 1/2 (internal)	15

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.8.1 Diphase operational test.

a. Monitor the Group Error detector. Verify that the requirements of 3.7.8.1 are met.

b. Monitor the DVOW and DOW Error detector. Verify that the requirements of 3.7.8.2 are met.

c. Set the appropriate Diphase B OW switch to the MAINT position.

d. Connect the oscillator at an output of 977 mV and 1 kHz to the associated Group AVOW input.

e. Monitor the Group Error detector as in step a.

f. Measure the AVOW output. Verify that the requirements of 3.7.8.3 are met.

g. Repeat step a through f for each remaining Diphase group.

4.7.8.2 Dipulse operational test.

a. Connect the Group Simulator to the first Dipulse module in the UUT. Set the Group Simulator and Dipulse CCA Group Rate to 2304 kb/s.

b. Connect the oscillator at an output of 977 mV and 1 kHz to the associated group AVOW input.

c. Monitor the group error detector. Verify that the requirements of 3.7.8.1 are met.

d. Measure the AVOW output. Verify that the requirements of 3.7.8.3 are met.

e. Repeat a through d for each remaining Dipulse group.

4.7.8.3 Bipolar operational test.

a. Connect the Group Simulator to the first Bipolar group in the UUT. Set the Group Simulator Group Rate 4915.2 kb/s.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currenversion before use.

b. Connect the oscillator at an output of 977 mV and 1 kHz to the associated Group AVOW input.

c. Monitor the group error detector. Verify that the requirements of 3.7.8.1 are met.

d. Measure the AVOW output. Verify that the requirements of 3.7.8.3 are met.

e. Repeat steps a through d for each remaining Bipolar group.

4.7.9 Environmental operational test. (See 3.7.9)

a. Connect the equipment as shown in figure 37 for Diphase groups and figure 38 for Dipulse and Bipolar groups.

b. Determine the type of module in each group of the UUT. Set the Group Simulator UUT and auxiliary unit switches as follows:

CCA Type	Function	UUT and Aux Unit Switches	Group Simulator
DIPHASE A DIPHASE B	GROUP RATE ORDERWIRE	12 (2304 kbps) SYS	12
DIPULSE	GROUP RATE TRF TRANS MI REC MI	12 (2304 kbps) NORM 1 (internal) 1 (internal)	12
BIPOLAR	GROUP RATE TRF TRANS MI REC MI	15 (4915.2 kbps) NORM 1/2 (internal) 1/2 (internal)	15

4.7.9.1 Dipulse and bipolar maintenance OW output check. (See 3.7.9.1)

a. With the Audio Oscillator set to -4dBm at 1.6 kHz, measure the MOW output with the Transmission Measuring Set.

b. Verify that the requirements of 3.7.9.1 are met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.7.9.2 Group data error check with OW test tone. (See 3.7.9.2)

a. With the Telephone Oscillator set to -4dBm at 1.6 kHz, monitor each of the Bipolar and Dipulse Group Error Detectors at the Group Simulator.

b. Verify that the requirements of 3.7.9.2 are met.

4.7.9.3 Diphase group error check. (See 3.7.9.3)

a. Monitor the Diphase Group Error Detector.

b. Verify that the requirements of 3.7.9.3 are met.

4.7.9.4 DVOW error check. (See 3.7.9.4)

a. Monitor the DVOW Error Detector.

b. Verify that the requirements of 3.7.9.4 are met.

4.7.9.5 DOW error check. (See 3.7.9.5)

a. Monitor the DOW Error Detector .

b. Verify that the requirements of 3.7.9.5 are met.

4.7.9.6 Alarm monitor check. (See 3.7.9.6)

a. Monitor the S, SA, LS alarm outputs on the UUT. Observe there is an open circuit indicated on each Simpson meter. (A short indicates alarm).

b. Verify that the requirements of 3.7.9.6 are met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.8 Service conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2 Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +63°C (+145°F). Maintain internal chamber temperature for 6 hours at +63°C (+145°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +63°C (+145°F) within a time period of one hour. Repeat the above steps two additional times for a total of three 12-hours cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +63°C (+145°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

4.8.1.2 Low temperature. (See 3.8.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C(-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation. (Para. II- 3.2 of Method 502.2). The chamber temperature in step 1 of Procedure II shall be adjusted to -31.7°C(-25°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III.

4.8.3 Low pressure (altitude). (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,000 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (20.6 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,000 fpm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.8.4 Dust. (See 3.8.4) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. In step 2 and step 5, the air velocity shall be 200 ±100 feet per minute. In step 5, the air temperature shall be adjusted to +63°(+145°F). The equipment shall be reoriented periodically to expose all its sides to the dust stream. The equipment shall be operated before step 1, during step 8 and during step 11.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C(+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall be no clogging or binding of moving mechanical parts.

4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs, etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Vibration and shock. (See 3.8.7)

4.8.7.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3, Procedure I, Category 8 (Ground Mobile). The vibration level shall be in accordance with figures 514.3-7 through 514.3-9 of MIL-STD-810D. The axes designated on the schedules are the vehicle axes. The orientation of the equipment relative to the vehicle, therefore, must be considered for proper direction of vibration input. Test time shall be 4 hours in each of three mutually perpendicular directions for a total of 12 hours. Prior to and following these tests, the equipment shall be operated in accordance with MIL-STD-810D general test performance guidance paragraph 4.5. The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

4.8.7.2 Shock.

4.8.7.2.1 Functional. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure I. The shock pulse shall be 20g peak for a nominal duration of 11 msec in accordance with figure 516.3-4.

4.8.7.2.2 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI.

4.9. EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government-approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First Article reliability test. First Article reliability testing shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC of MIL-STD-781C.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. Chamber temperature: 40°C ±5°C (95°F to 113°F).

b. Duty cycle shall be as follows: Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24-hour cycle.

c. Vibration: Sinusoidal, 2.2g ±10% peak acceleration at non-resonance frequency between 20 and 60 Hz measured at equipment mounting points. Apply for 10 minutes during each hour of testing.

4.10.4 Reliability test method. The detailed test procedures, and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor-prepared, government-approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.8.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimension of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

d. Location of panel-mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

f. In order to prove interchangeability between the different modules, perform a Group A test with all Diphase Modules installed, then all Dipulse Modules installed and finally all Bipolar Modules installed in the unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings listed in 3.12.

4.13 Visual and mechanical. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (Systems safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System safety engineering.)

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-.116.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Material inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table XII, "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedure shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

TABLE XII. Packing inspection provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification incorrectly applied, omitted	Visual
109	Marking omitted, incorrect or illegible	Visual

4.15 Serial numbers. The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and CCA.

4.16 Nuclear tests. (See 3.16) Applicable tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

4.17 TEMPEST tests. (See 3.17) TEMPEST tests shall be performed on the equipment in accordance with CR-CX-0174-001.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, tables I through VII, coded as follows:

5.1.1 Level A. 3Q-1-00-OO-NR-X-ED.

5.1.2 Level B. 2M-1-00-OO-NR-X-ED.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The Modem, Digital Data, MD-1026(P)/G provides for cable transmission by accepting balanced non-return-to-zero (NRZ) group signals which are converted to conditioned diphase, dipulse, or bipolar signals and transmitted over CX-11230 cable. The unit, commonly known is the Group Modem or GM, performs the inverse functions for the conditioned diphase, dipulse, of bipolar signals received from the cable. The GM is designed for use in shelters and vans and will be mounted in an equipment case. The GM may contain the following module types in different combinations:

a. Conditioned Diphase modules.

b. Dipulse module.

c. Bipolar module.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

c. When first article rough handling tests are not required.

d. Quantity of first article units required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirements.

i. MIL-Q-9858.

j. Length of cables (if any required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unt.

l. If the equipment is procured with any module types not included, it may be necessary to GFE those modules types to insure that all the requirements of this specification are satisfied.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. (See 4.5.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. Initial shipments of end items will be contingent upon successful completion of all first article tests.

6.5 Location of operational inspection. It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table VIII.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject terms (key word) list.

Balanced
Bipolar
Conditioned Diphase
Diphase
Digital Group Multiplex
Orderwire
Group Modem (GM)
Modem
Modem, Digital Data
Modem, Digital Data, MD-1026(P)/G

CUSTODIAN
ARMY-CR

Preparing Activity
ARMY-CR
Project No. 5820-A825

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

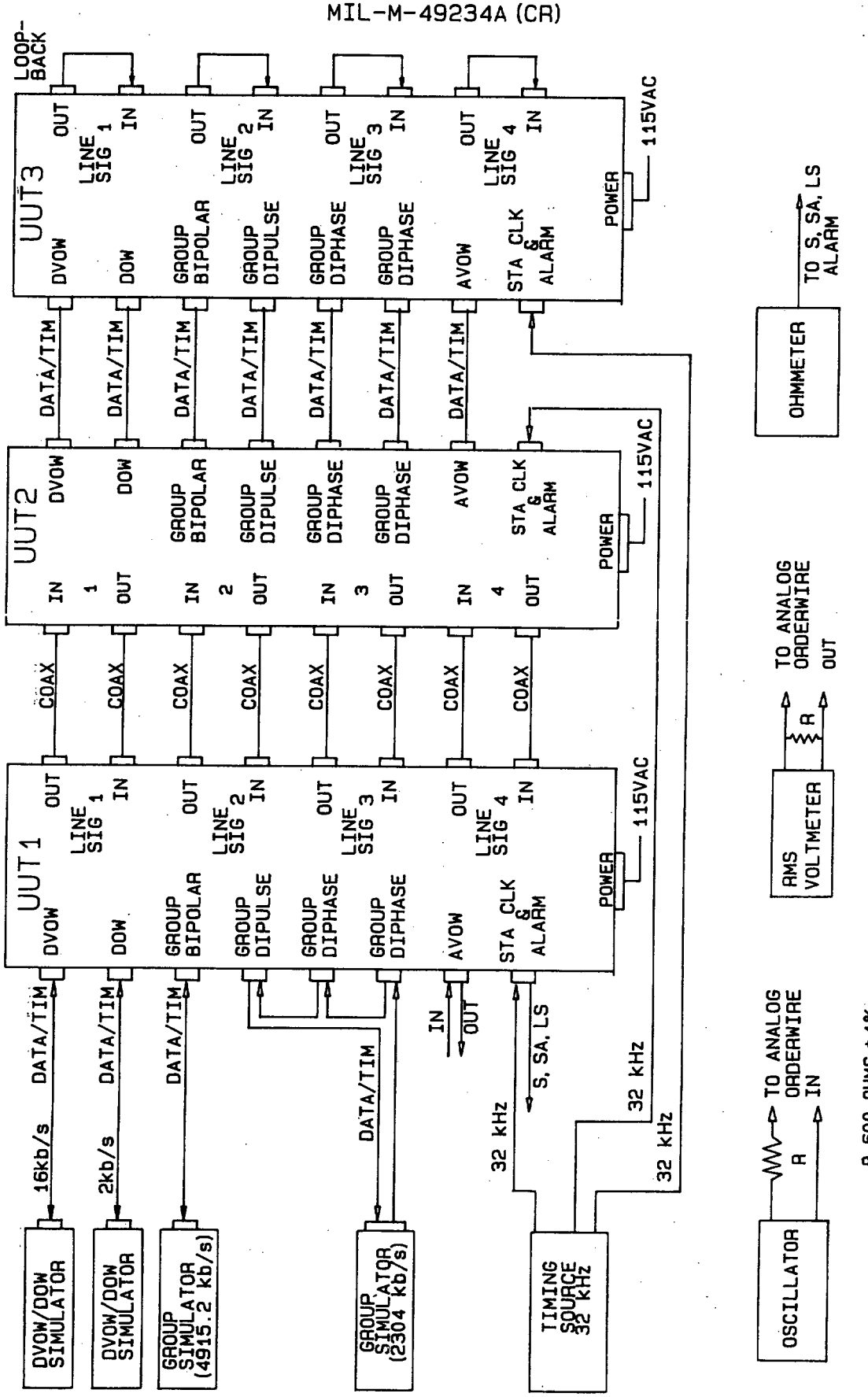
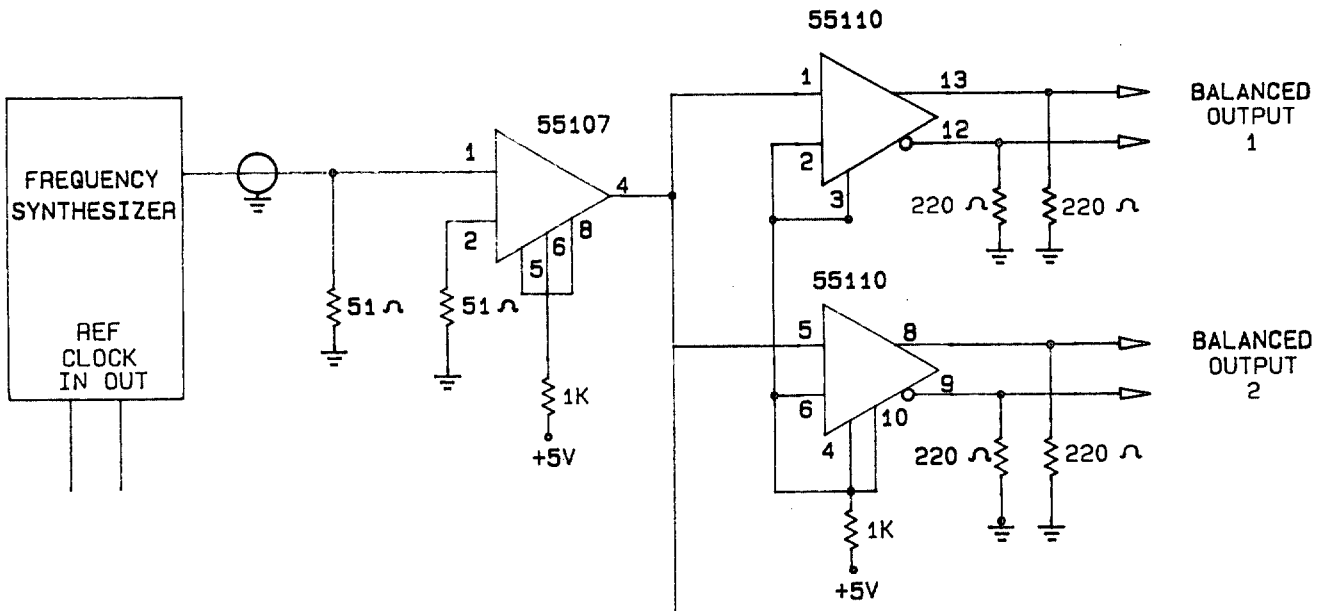


FIGURE 1. Burn-in.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.



GROUP RATE	SYNTHESIZER FREQUENCY
1	72 KHz
2	128 KHz
3	144 KHz
4	256 KHz
5	288 KHz
6	512 KHz
7	576 KHz
8	1024 KHz
9	1152 KHz
10	1536 KHz
11	2048 KHz
12	2304 KHz
13	4096 KHz
14	4608 KHz
15	4915.2 KHz

FIGURE 2. Alternate timing source.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

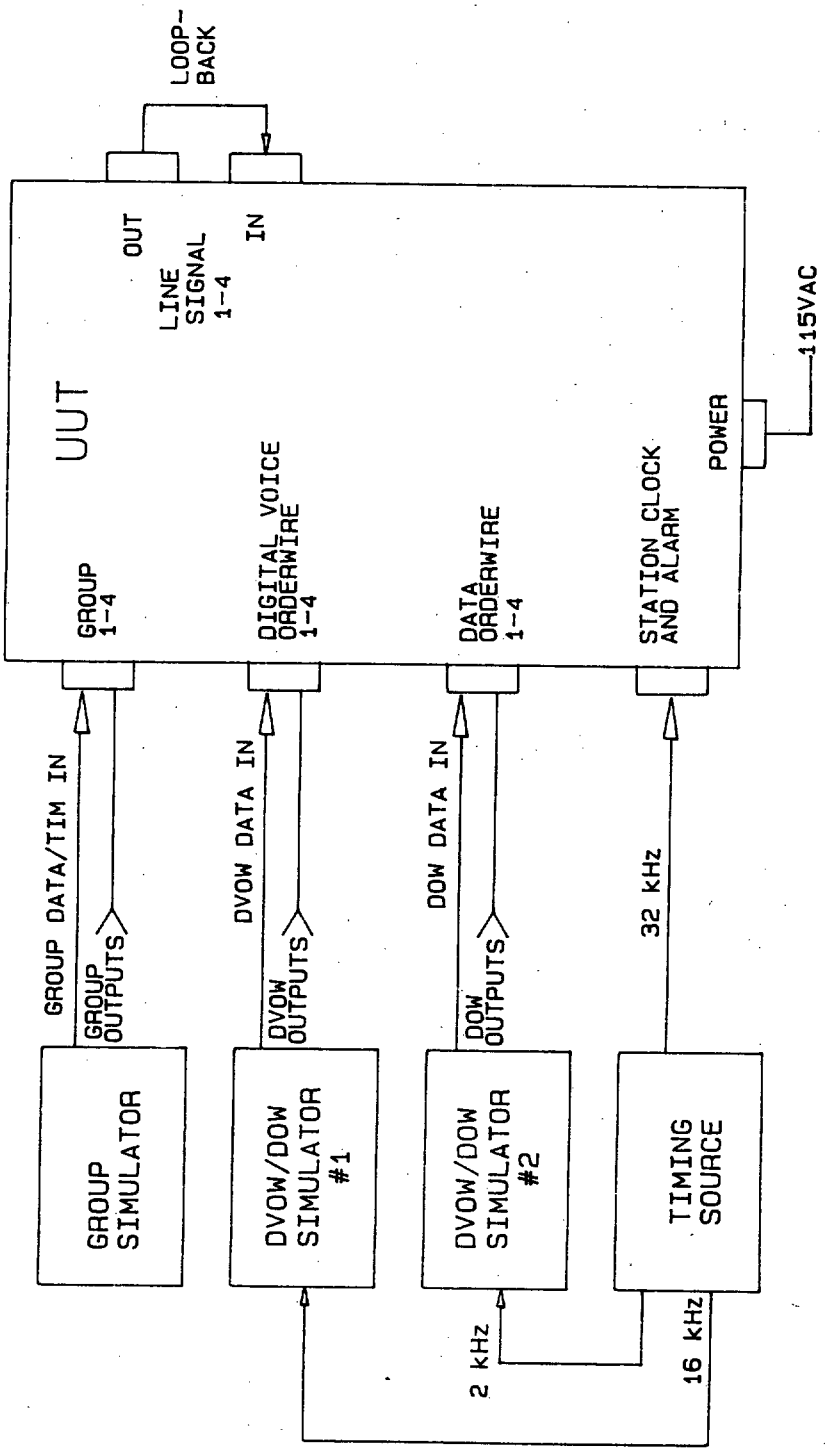


FIGURE 3. Line-to-line voltage levels.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

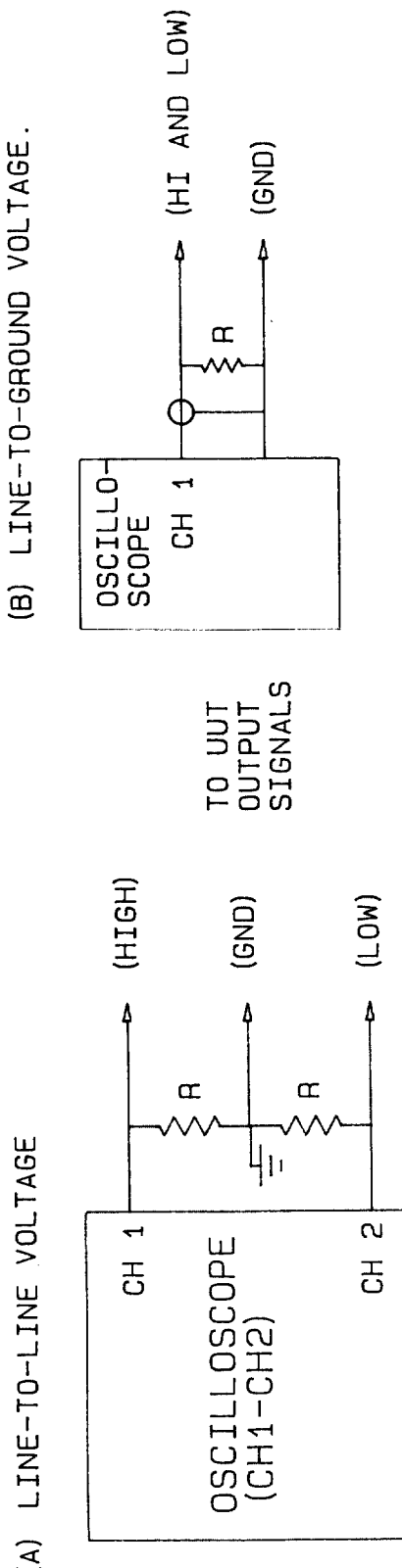


FIGURE 4. Oscilloscope connections: line-to-line and line-to-ground voltage.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

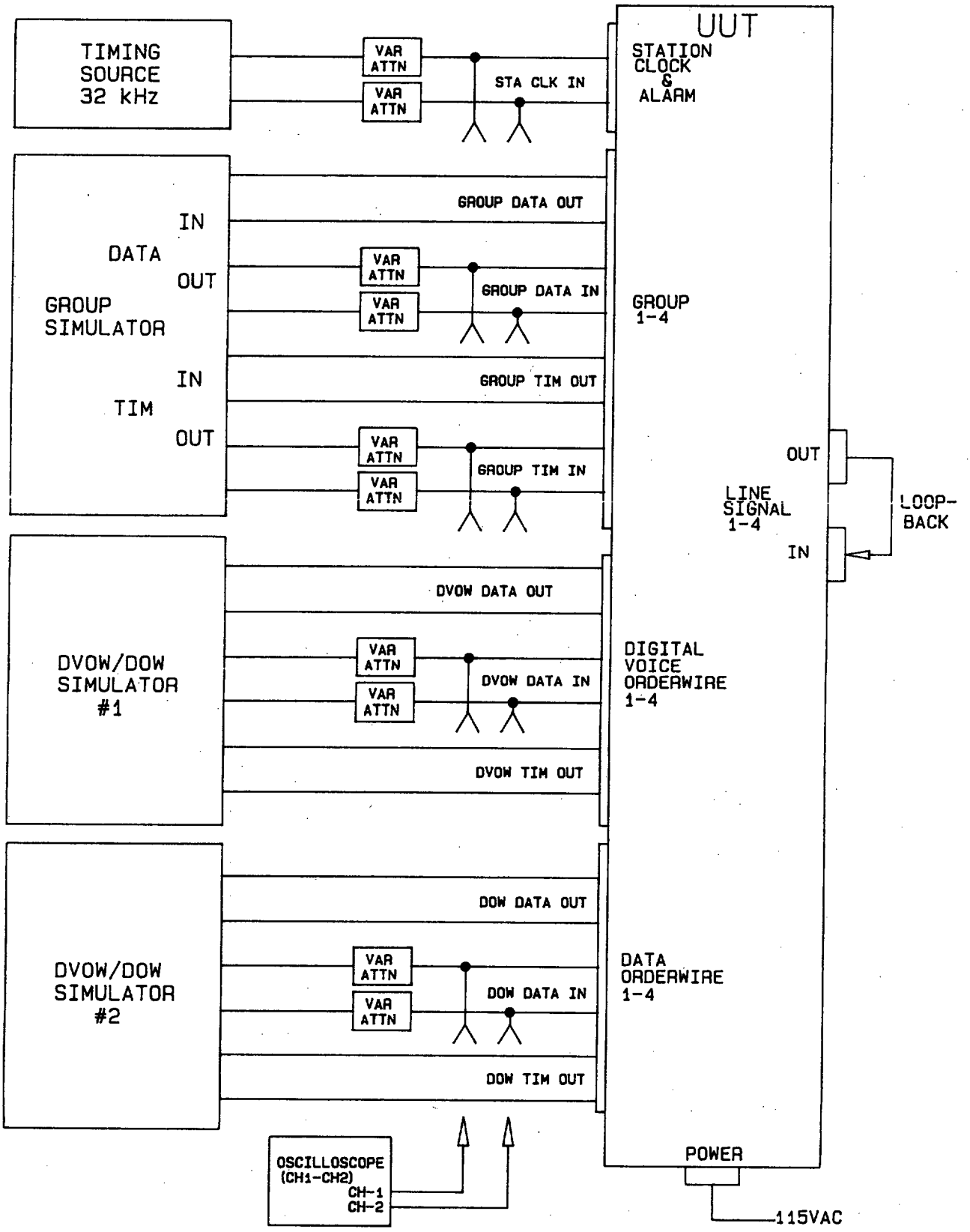


FIGURE 5. Receiver sensitivity.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

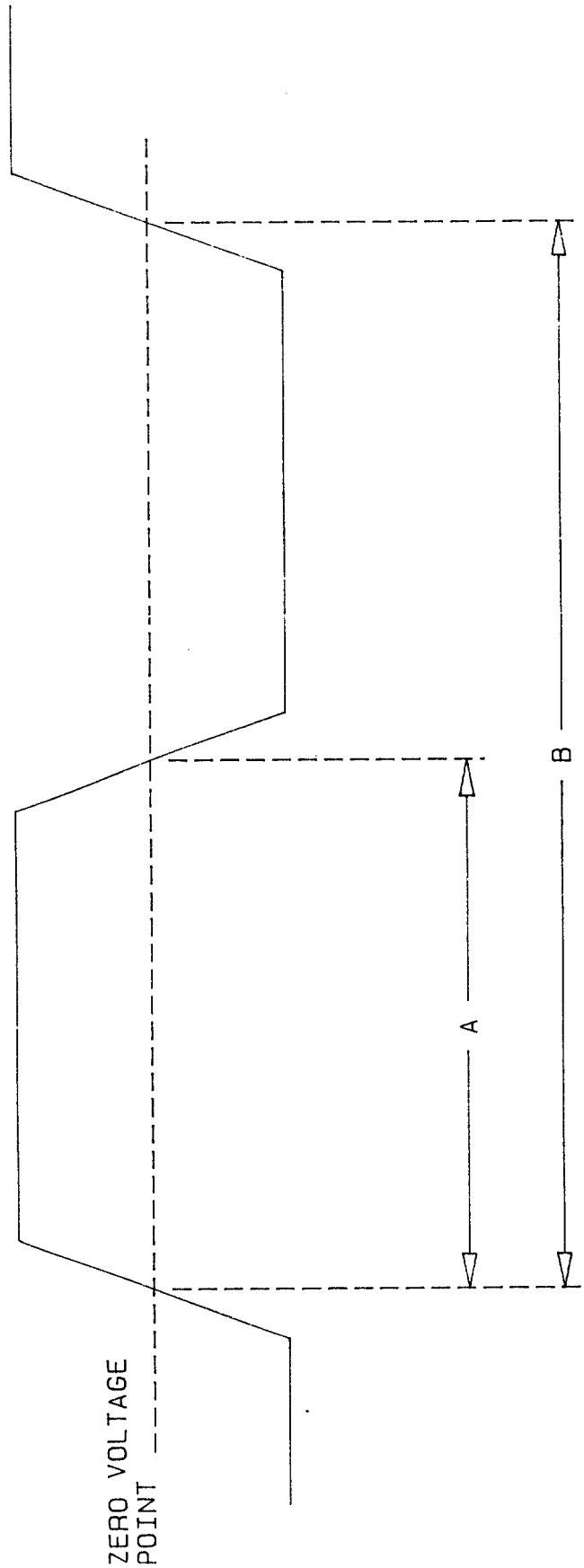


FIGURE 6. Clock asymmetry (clockwidth).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

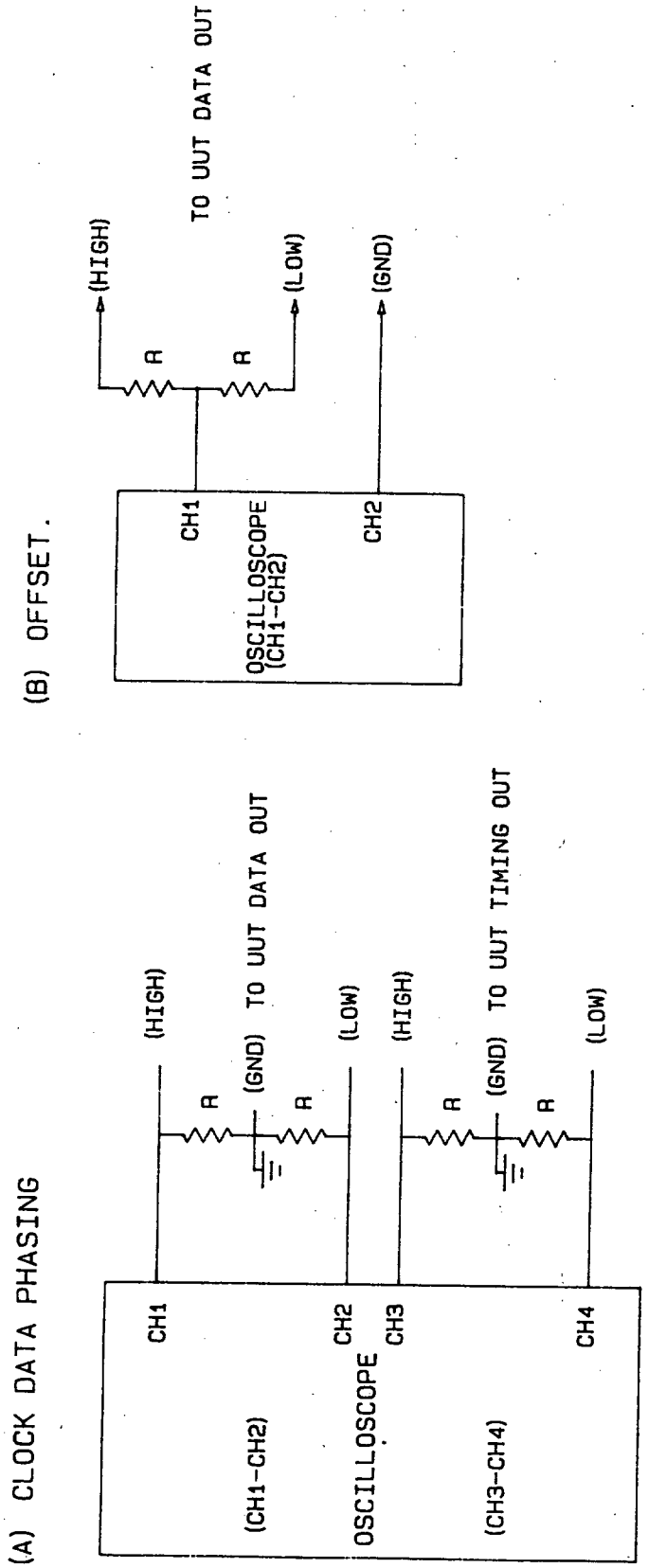


FIGURE 7. Oscilloscope connections: clock/data phasing; offset.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

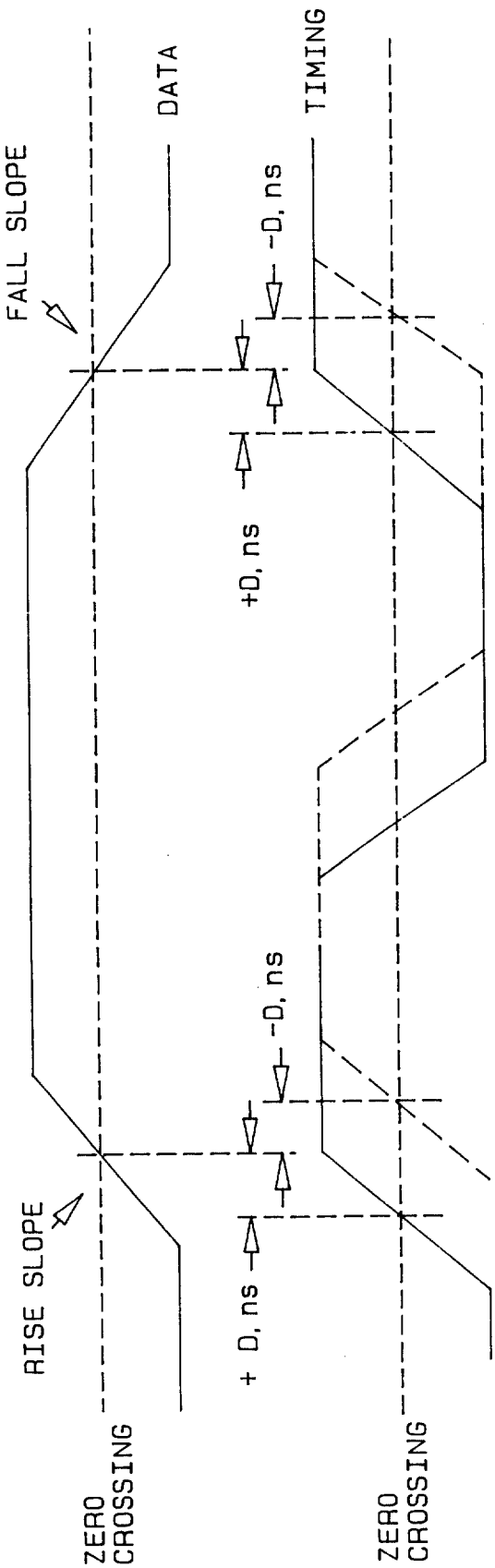


FIGURE 8. Delay measurement (clock/data phasing).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

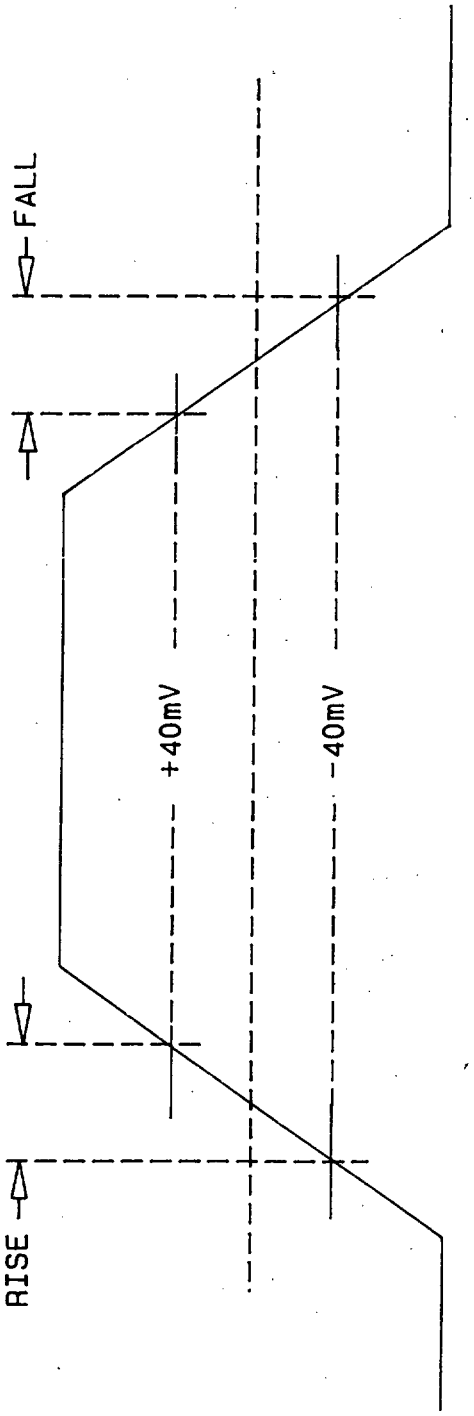


FIGURE 9. Rise and fall slopes (slew rate).

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

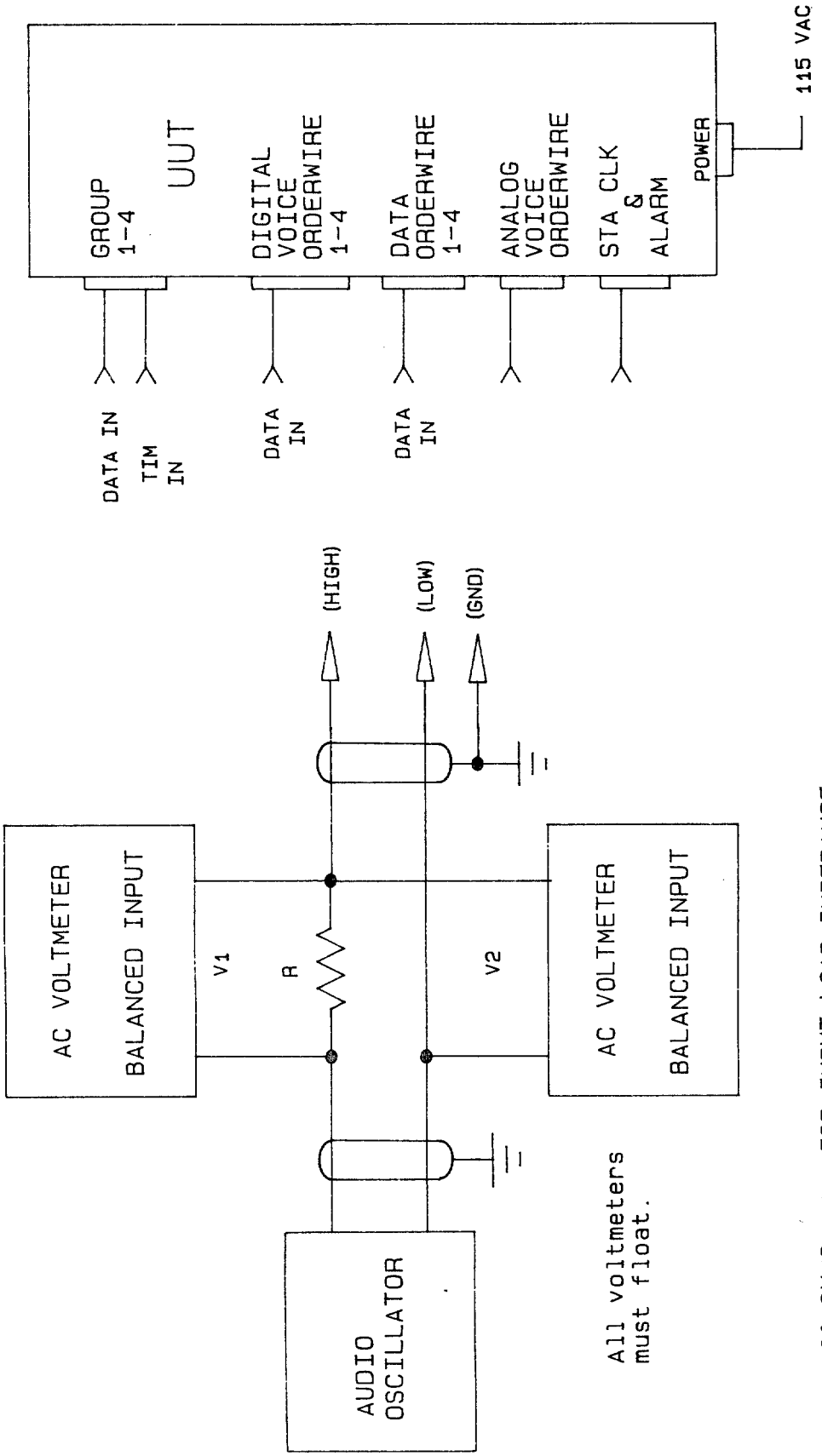


FIGURE 10. Input impedance.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

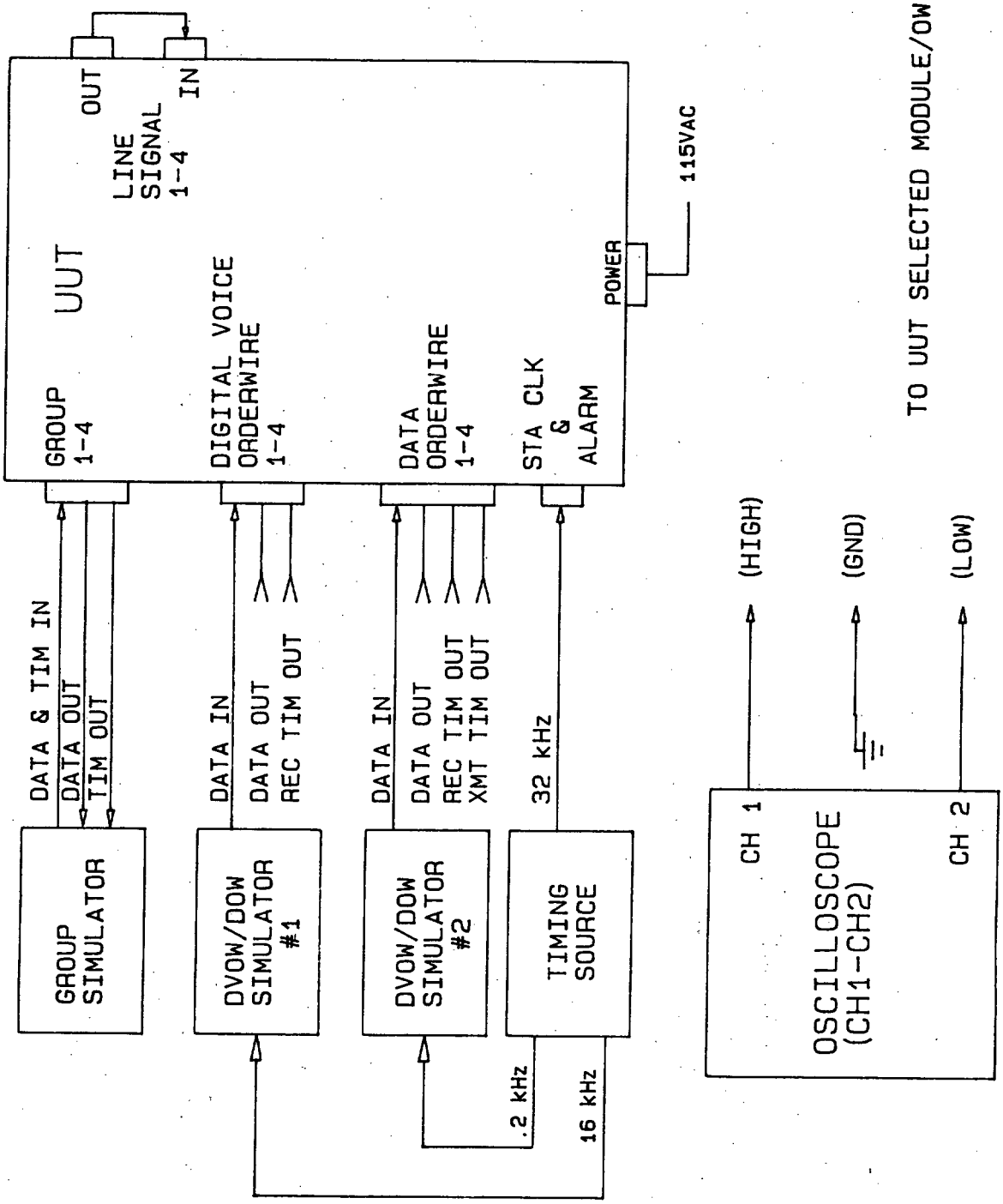
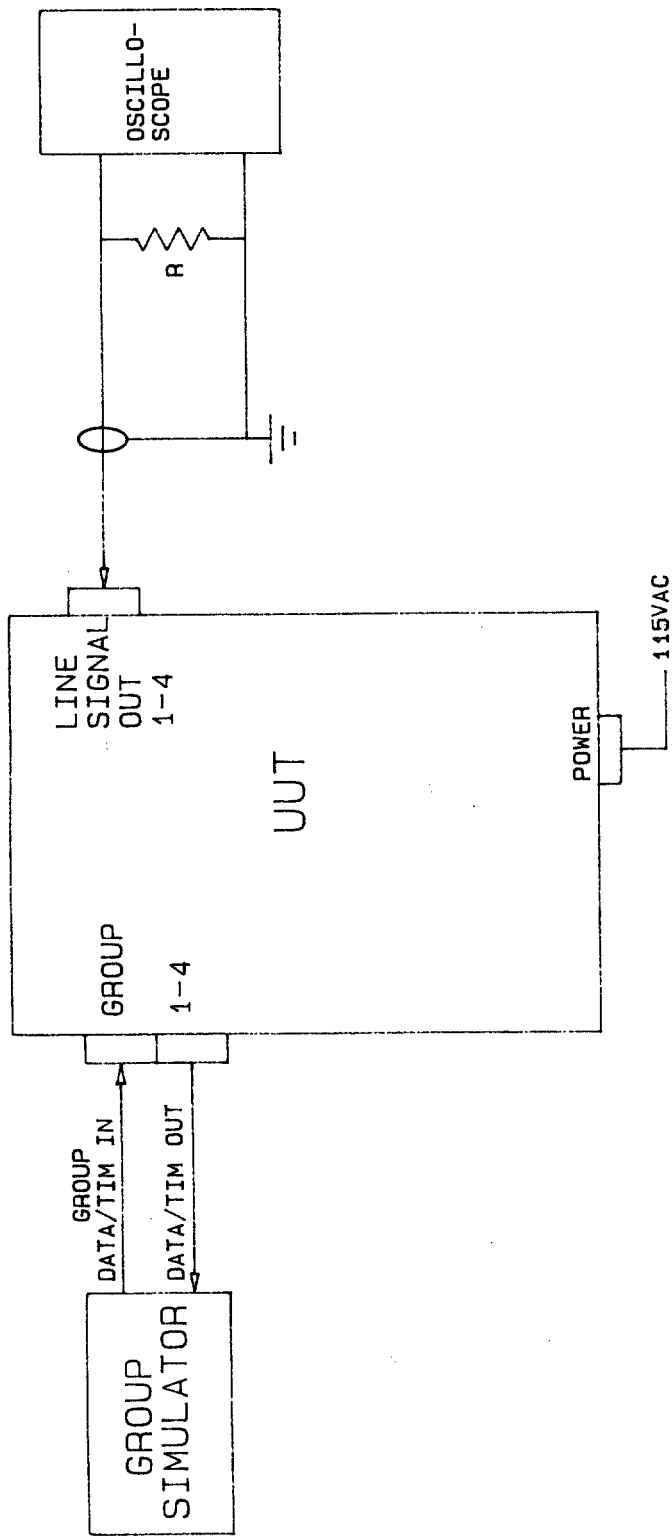


FIGURE 11. Line-to-line voltage levels, unloaded output.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.



R 58 OHMS ±1%

FIGURE 12. Output voltage level.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

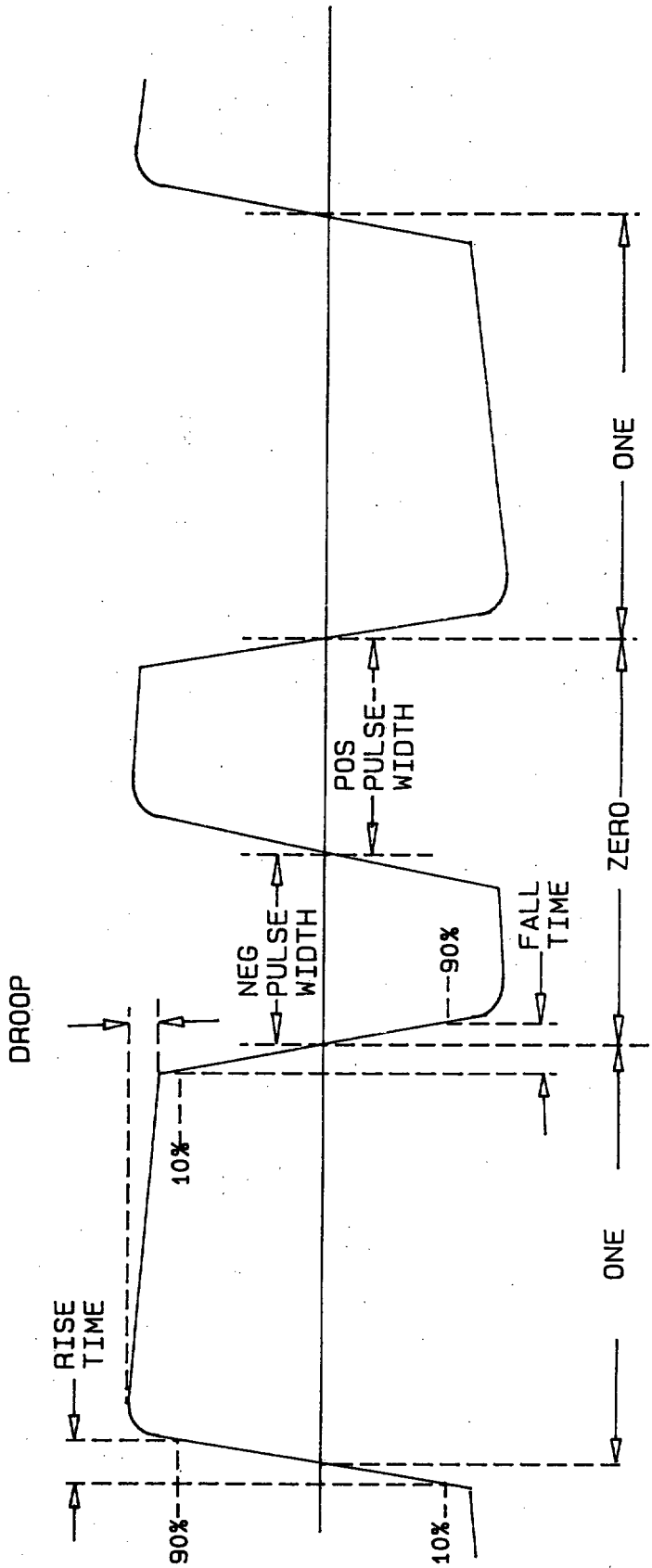


FIGURE 13. Diphase waveform (digital zero).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

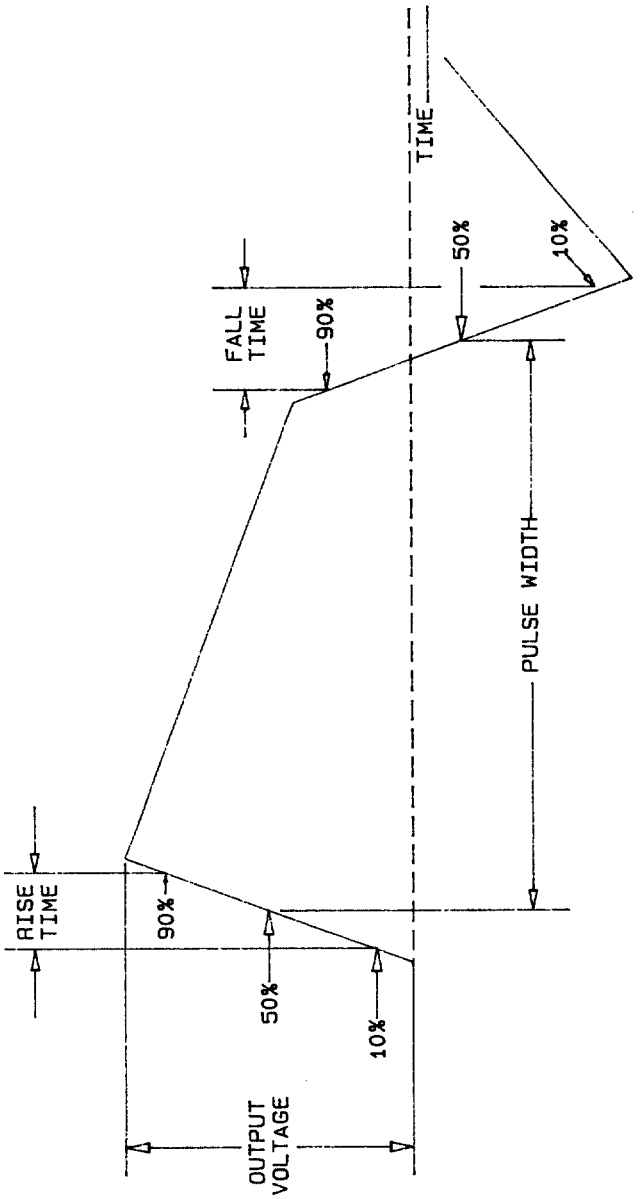


FIGURE 14. Dipulse waveform.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

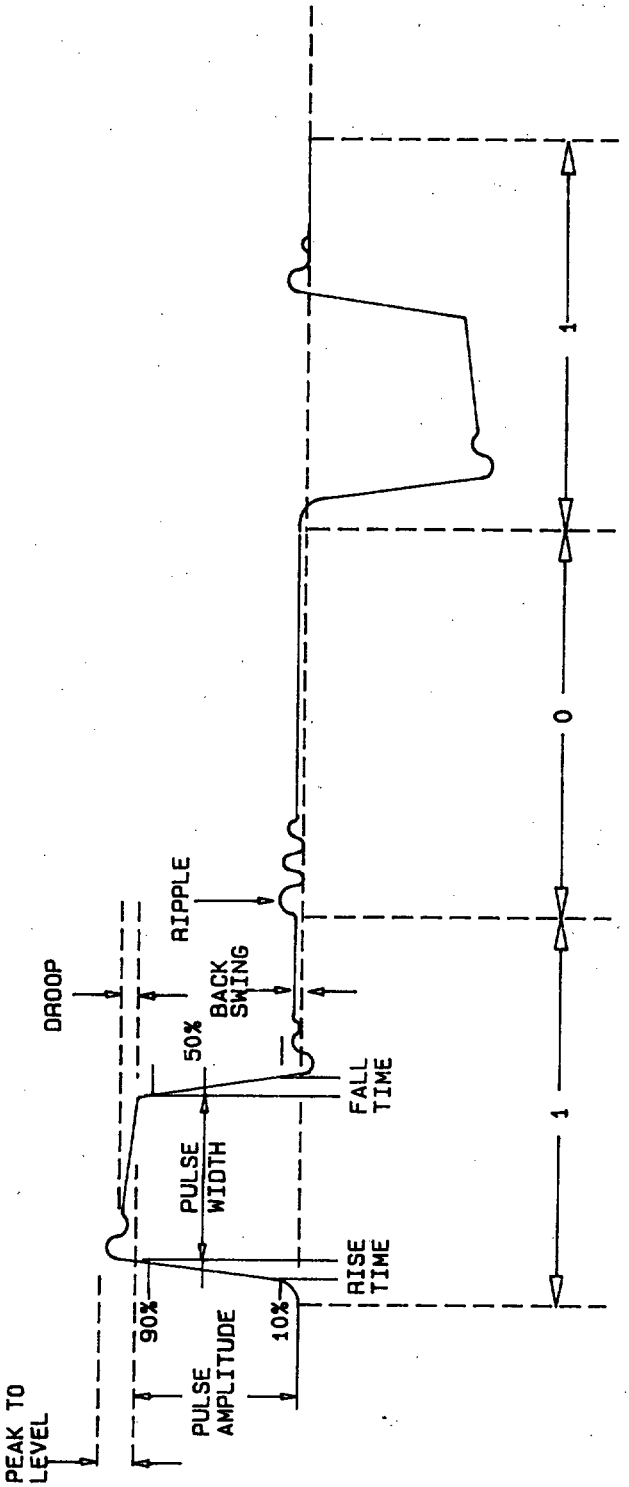
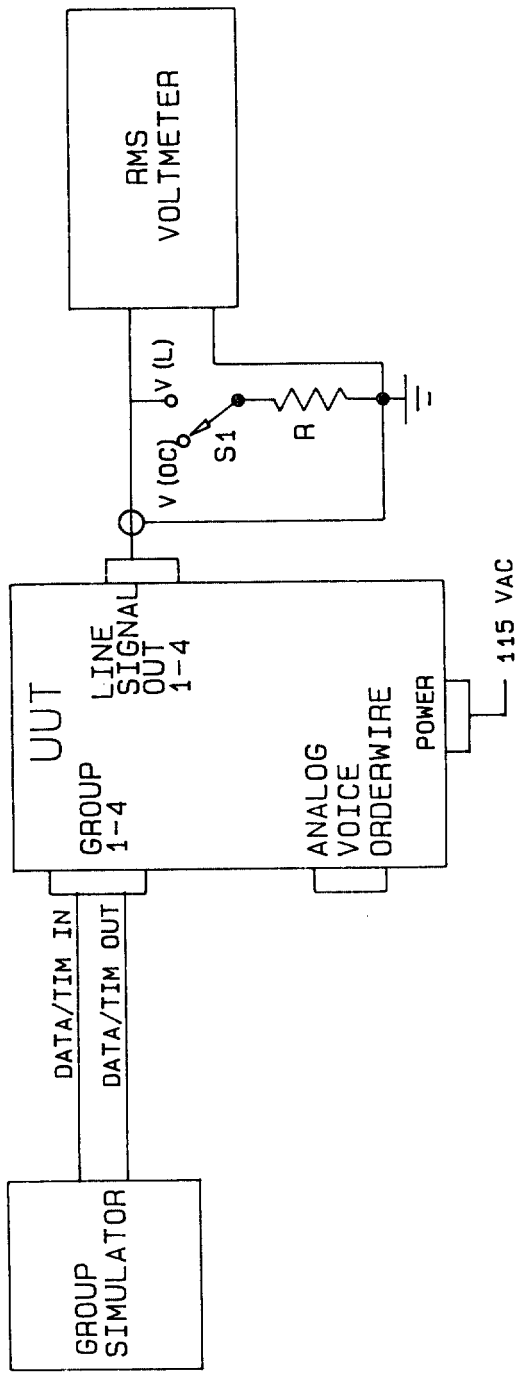


FIGURE 15. Bipolar waveform.

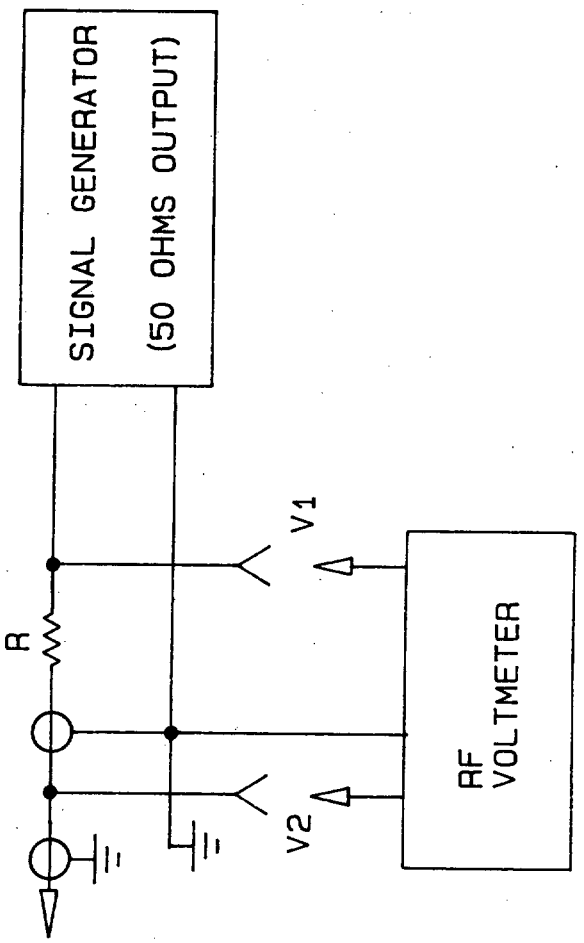
Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.



R 58 OHMS ±1%

FIGURE 16. Diphase and bipolar cable-side output impedance.

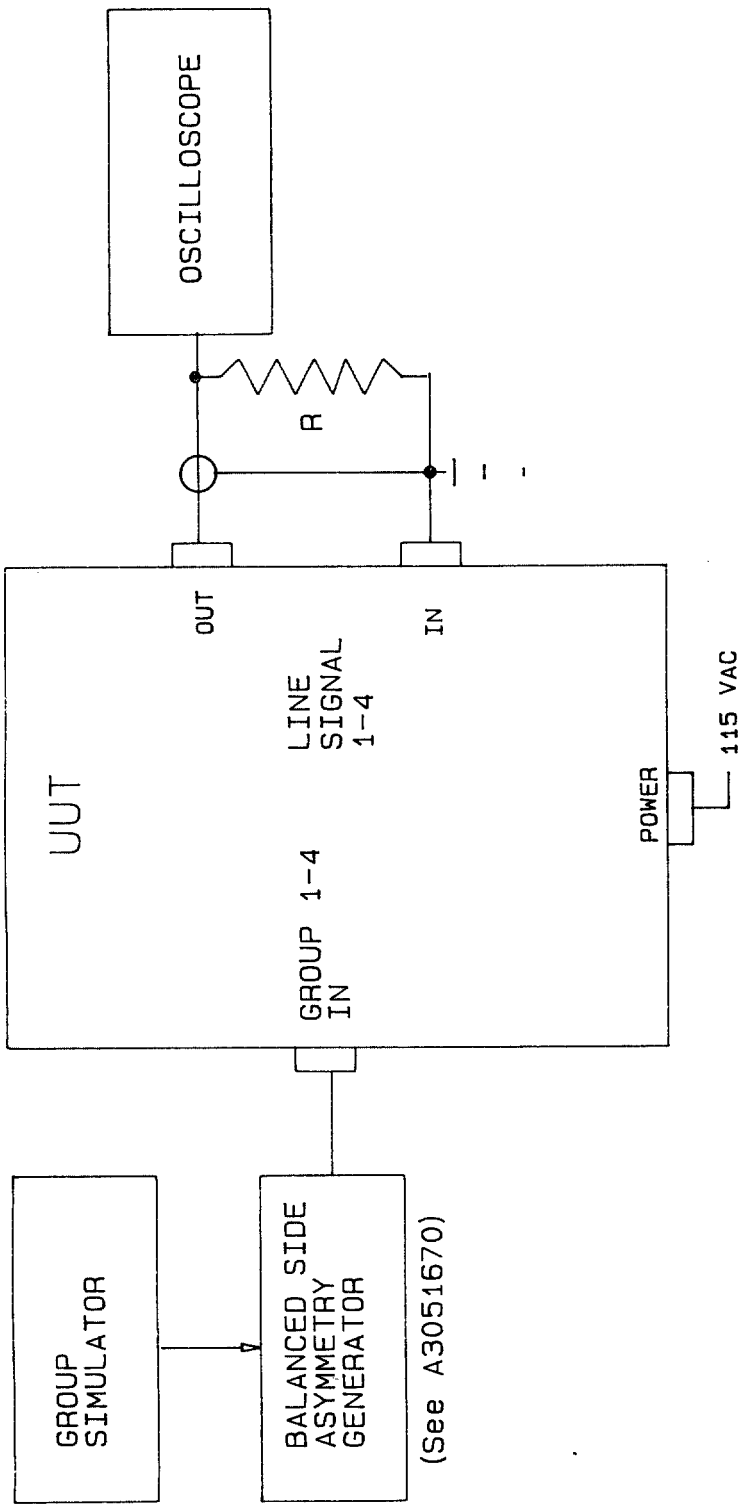
Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.



58 OHMS ± 1%

FIGURE 17. Diphase and bipolar input impedance.

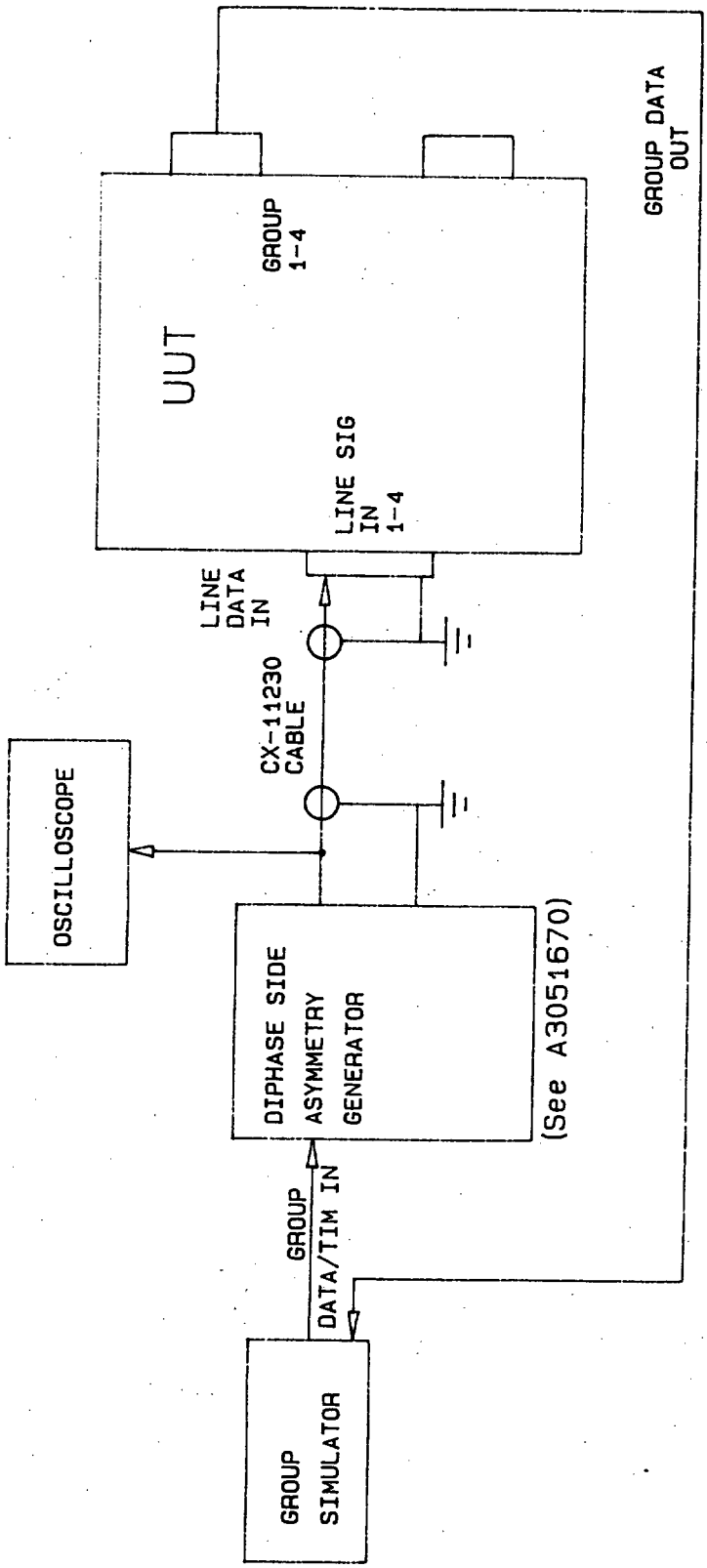
Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.



R 58 OHMS ±1%

FIGURE 18. Transmit asymmetry.

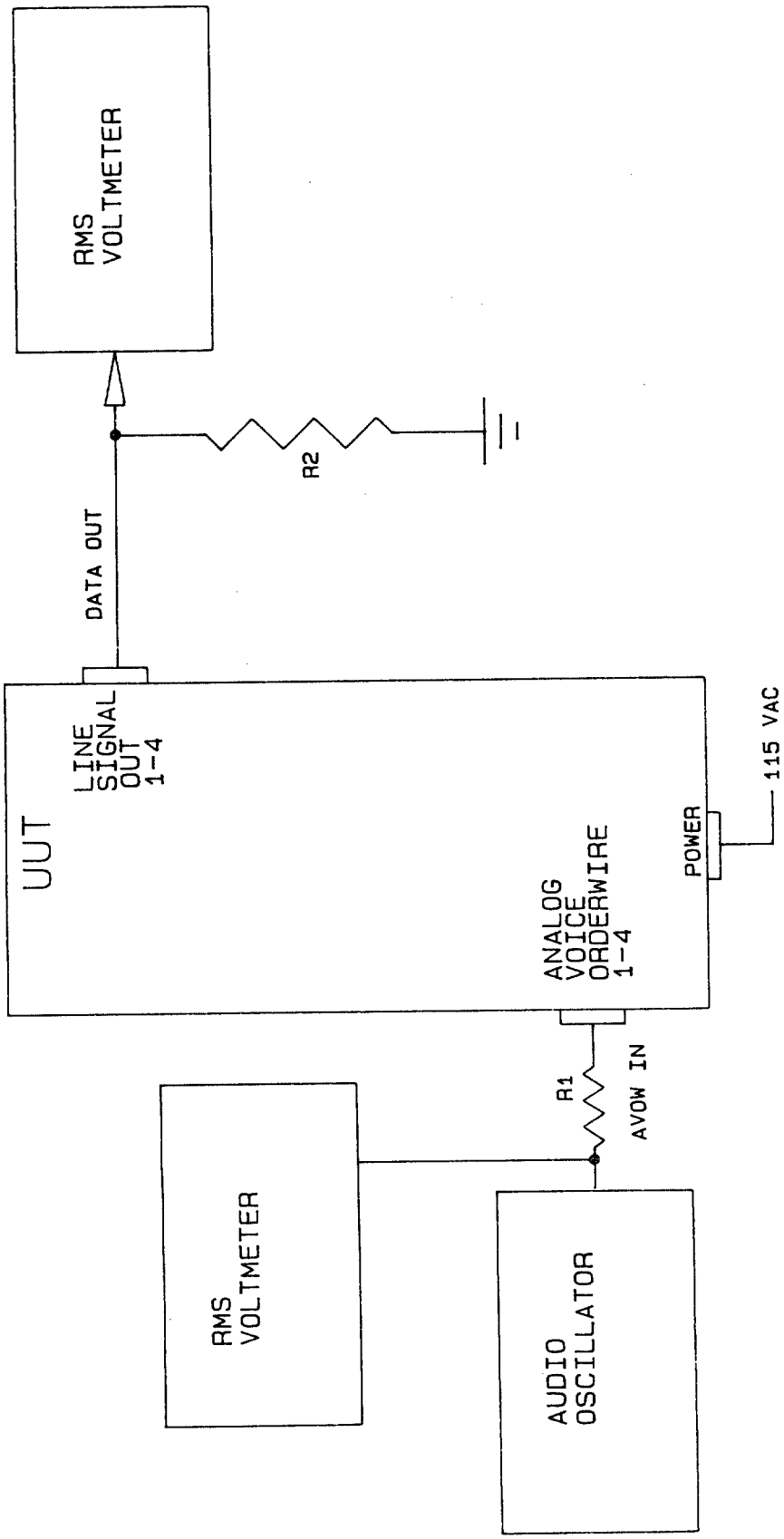
Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.



MAINT		SYS	
RATE	CABLE LENGTH	RATE	CABLE LENGTH
72	0 MILES AND 2 MILES +4dB	256	0 MILES AND 2 MILES +4dB
144	0 MILES AND 2 MILES +4dB	576	2 MILES +4dB
		1024 & 2304	1 MILE +4dB
		4096	1/2 MILE +4dB
		4608	0 MILES AND 1/2 MILE +4dB

FIGURE 19. Receiver asymmetry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.



R1 600 OHMS ±1%
R2 58 OHMS ±1%

FIGURE 20. Diphase maintenance orderwire cable-side output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

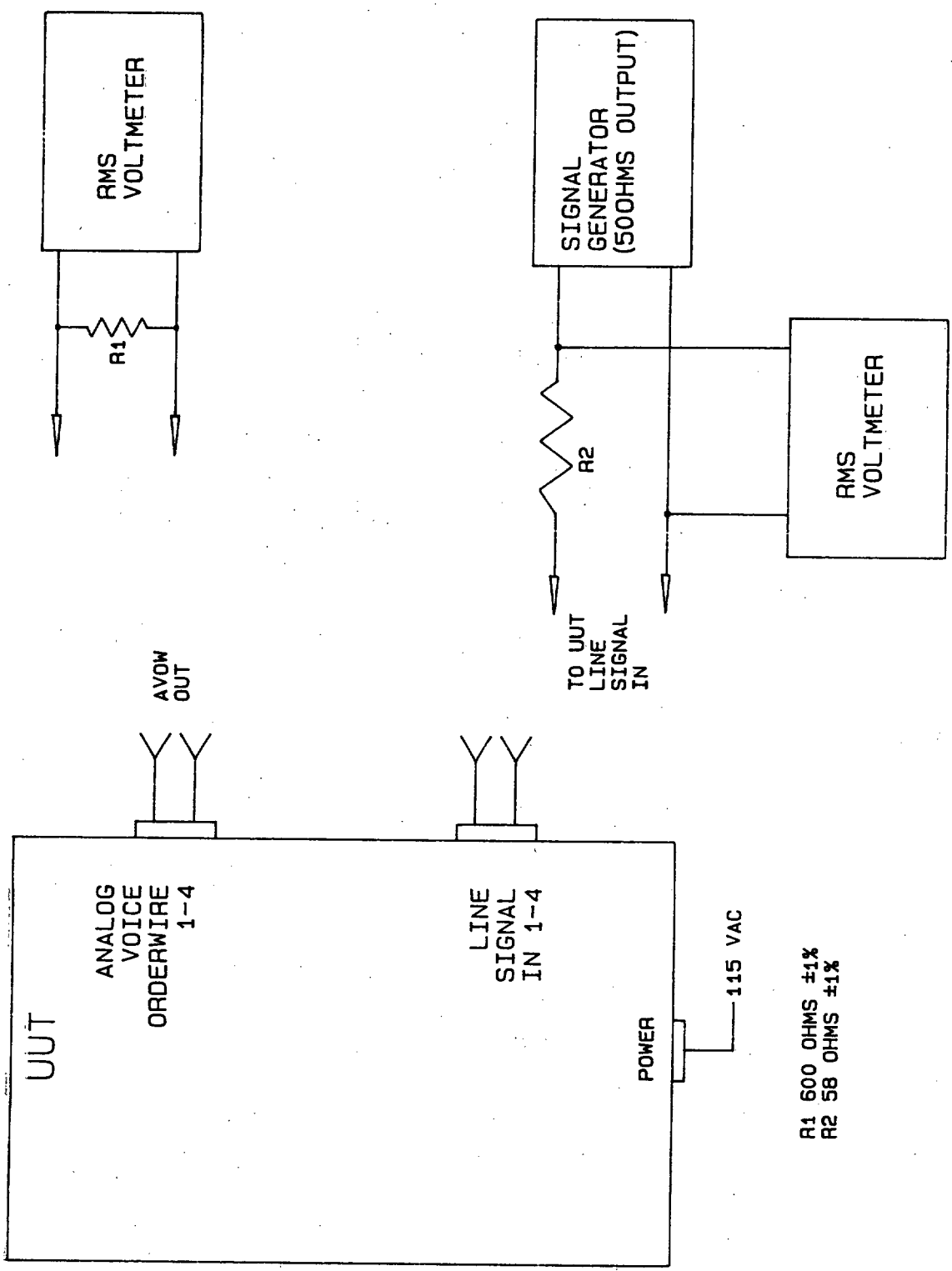


FIGURE 21. Diphase maintenance orderwire equipment-side output.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

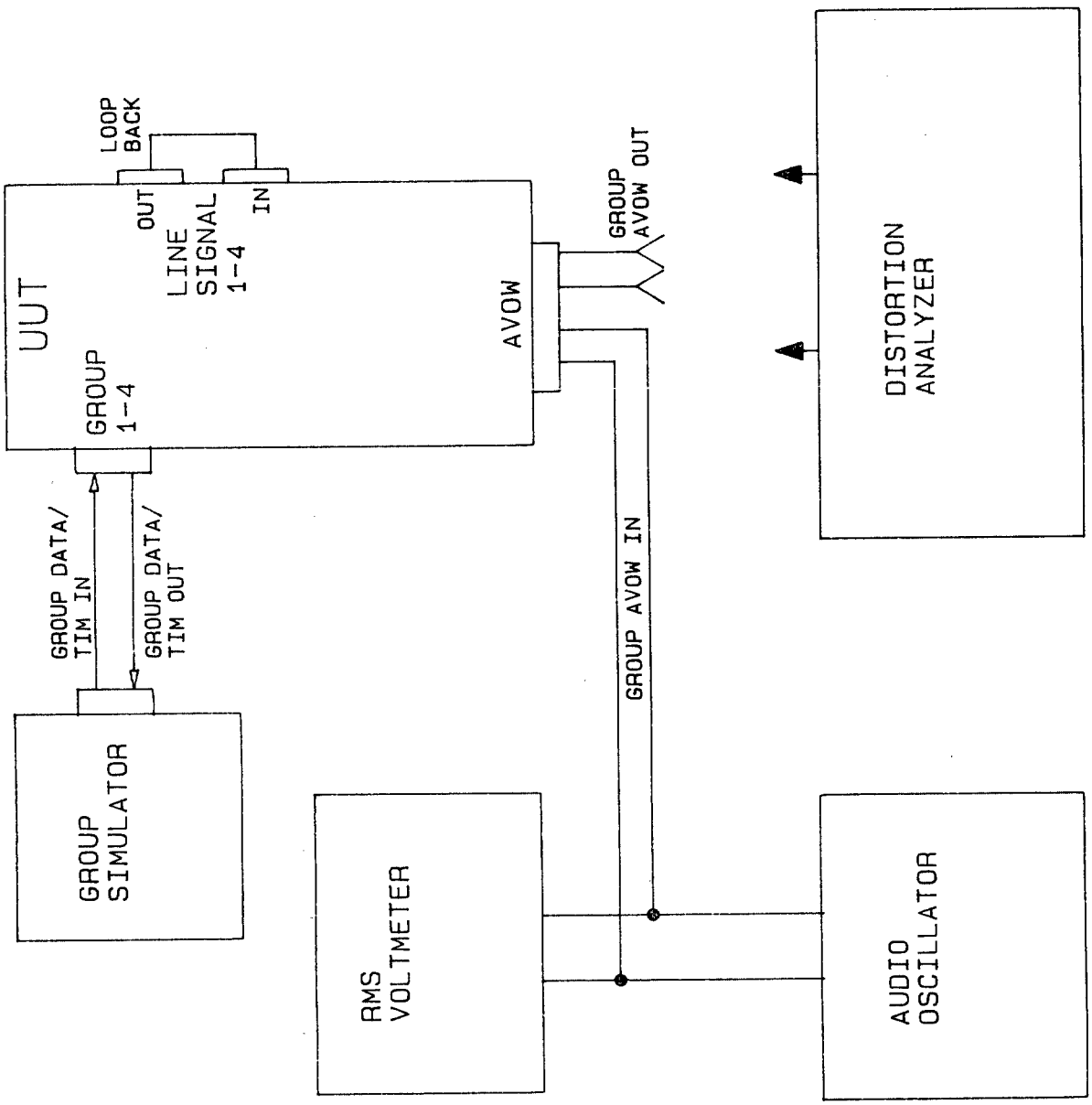
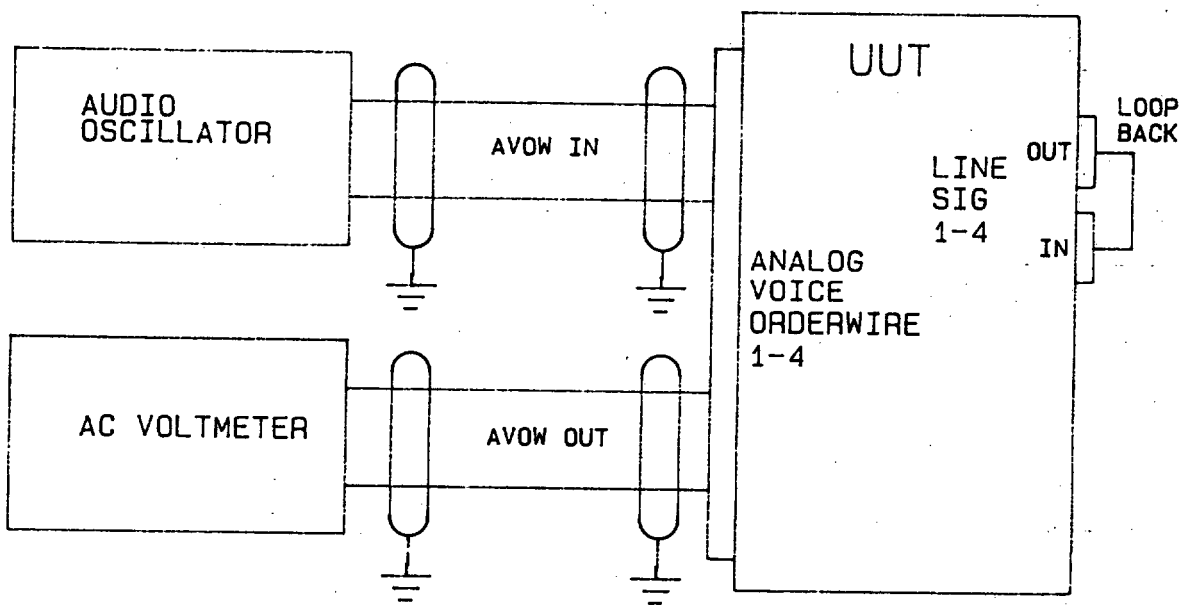


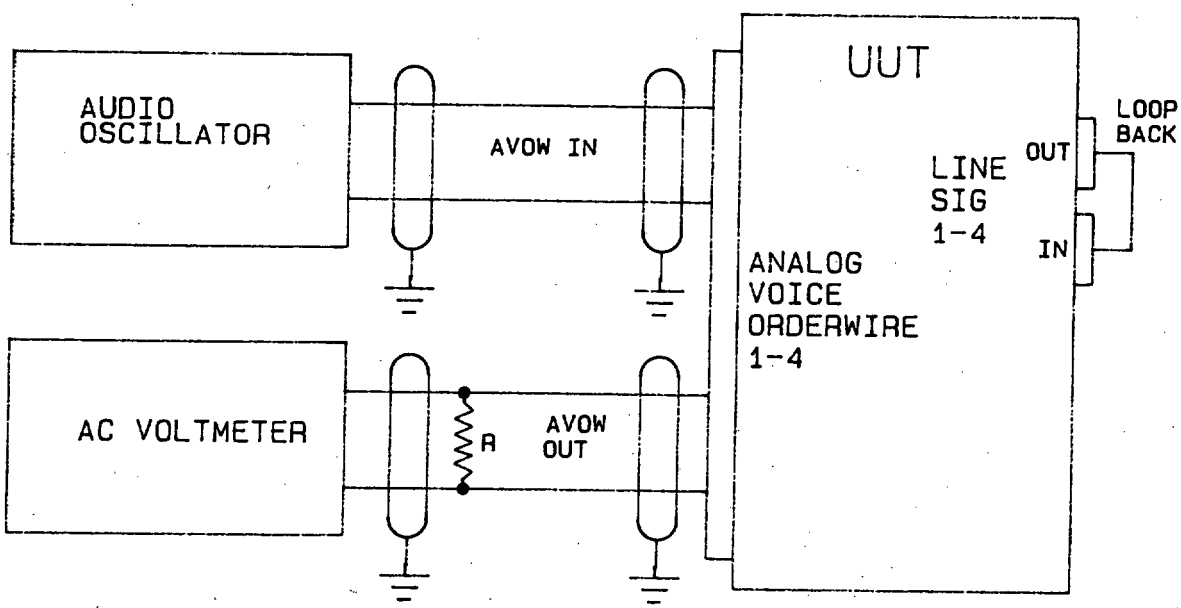
FIGURE 22. Signal-to-distortion ratio.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

(A) .MEASURING V1



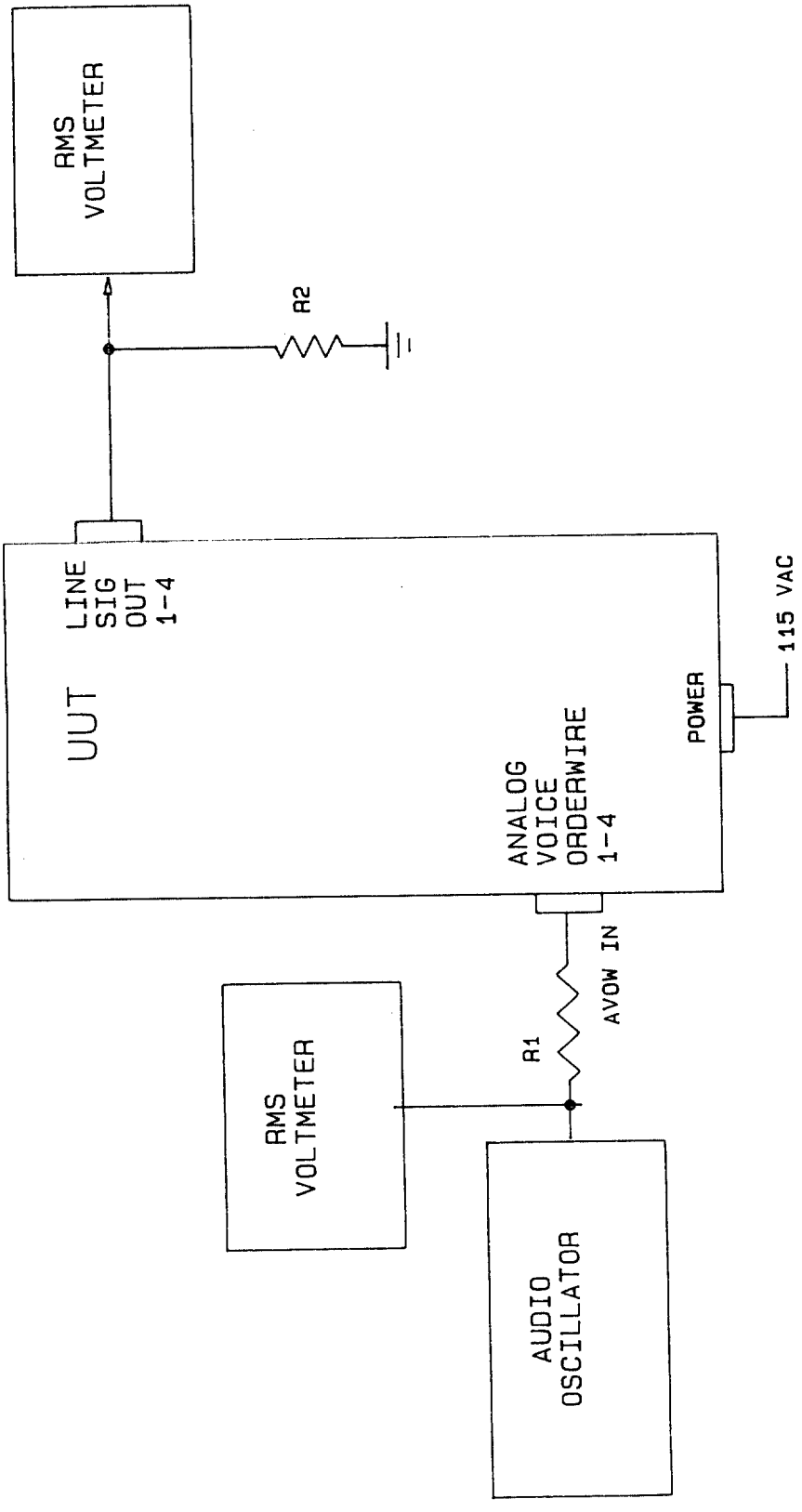
(B) .MEASURING V2



R 600 OHMS ±1%

FIGURE 23. Equipment-side maintenance orderwire output impedance.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.



R1 600 OHMS ±1%
R2 880 OHMS ±1%

FIGURE 24. Dipulse and bipolar maintenance orderwire cable-side output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

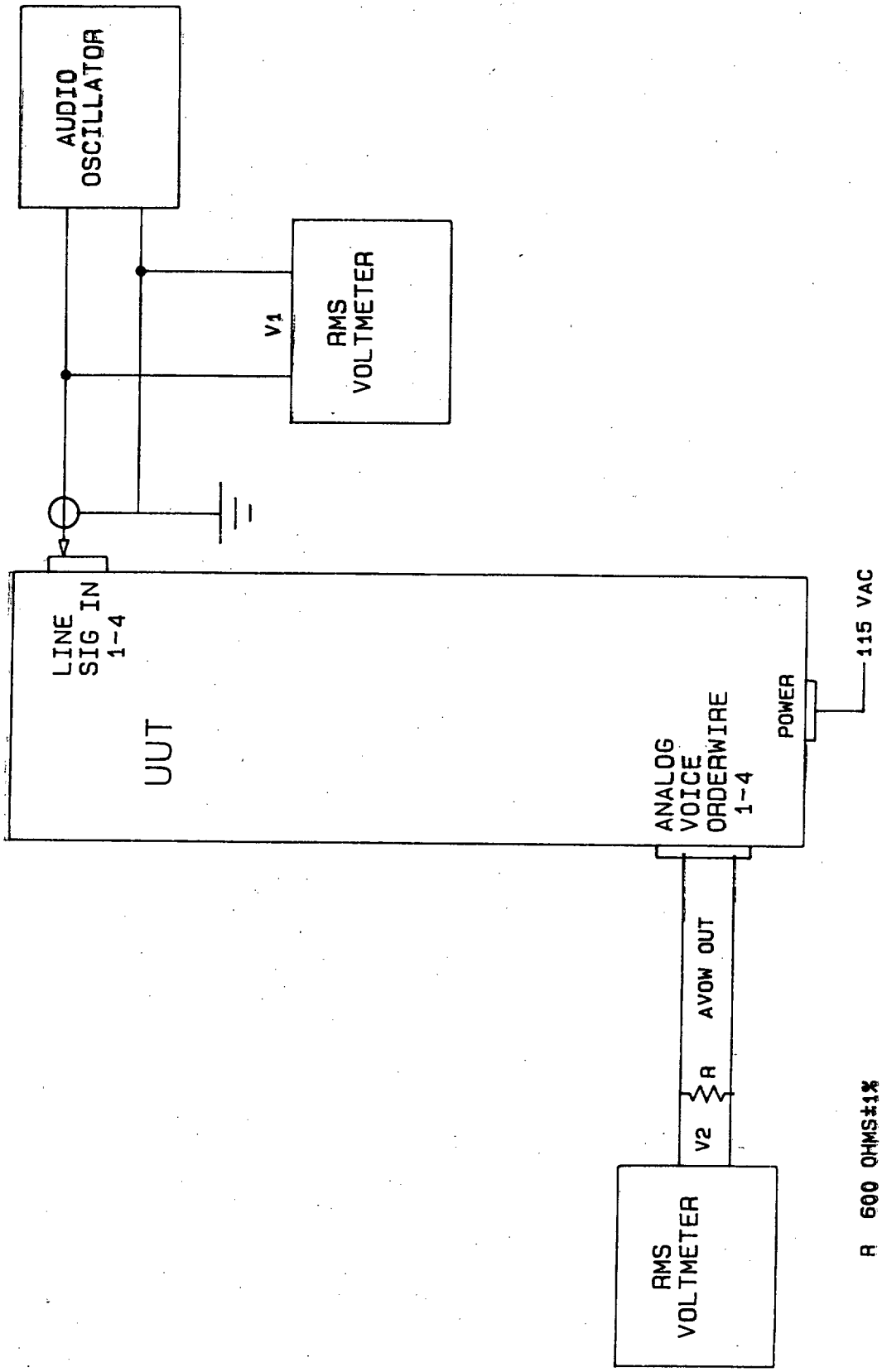


FIGURE 25. Dipulse and bipolar maintenance orderwire equipment-side output.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

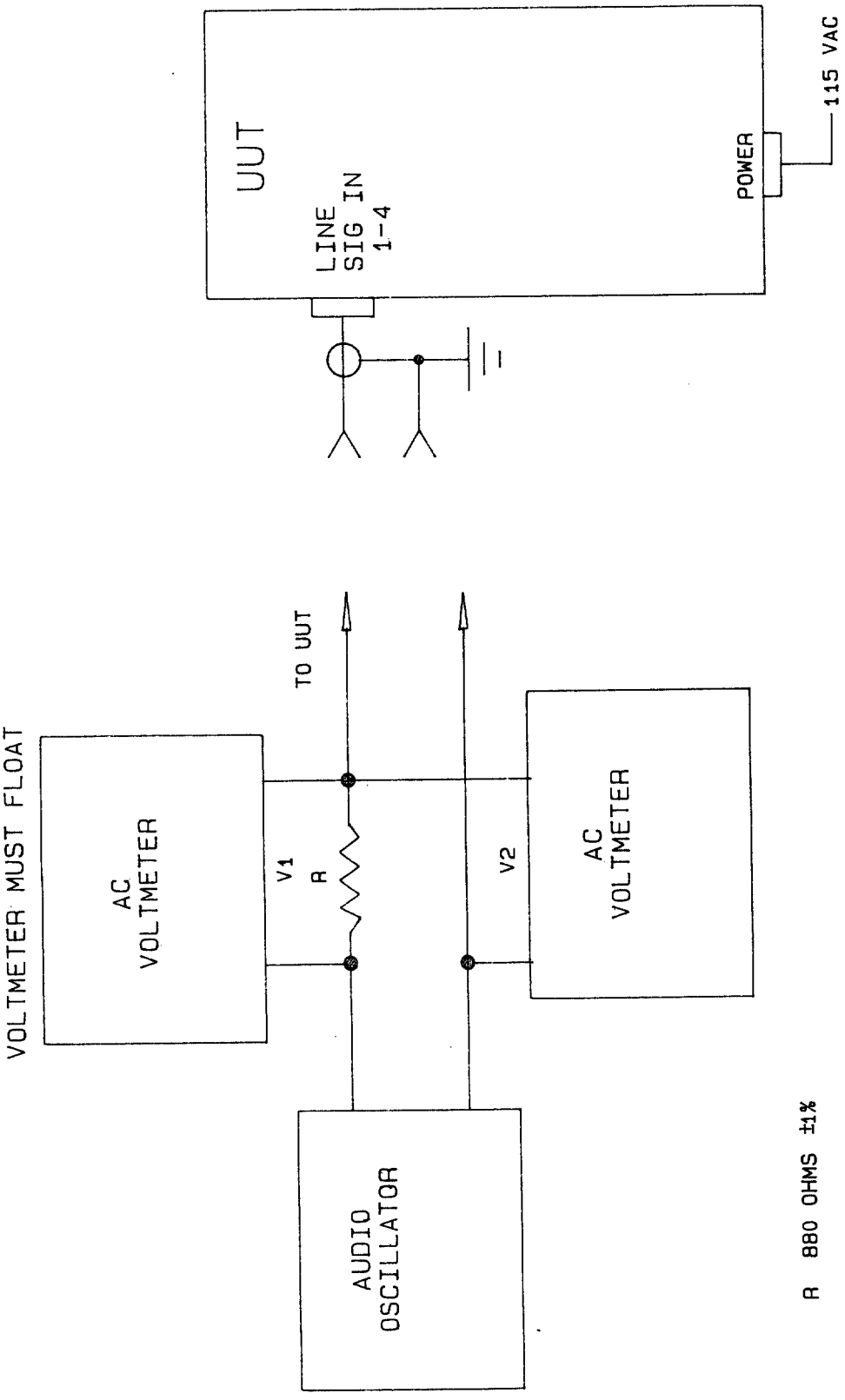


FIGURE 26. Dipulse and bipolar maintenance orderwire cable-side input impedance.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the currentversion before use.

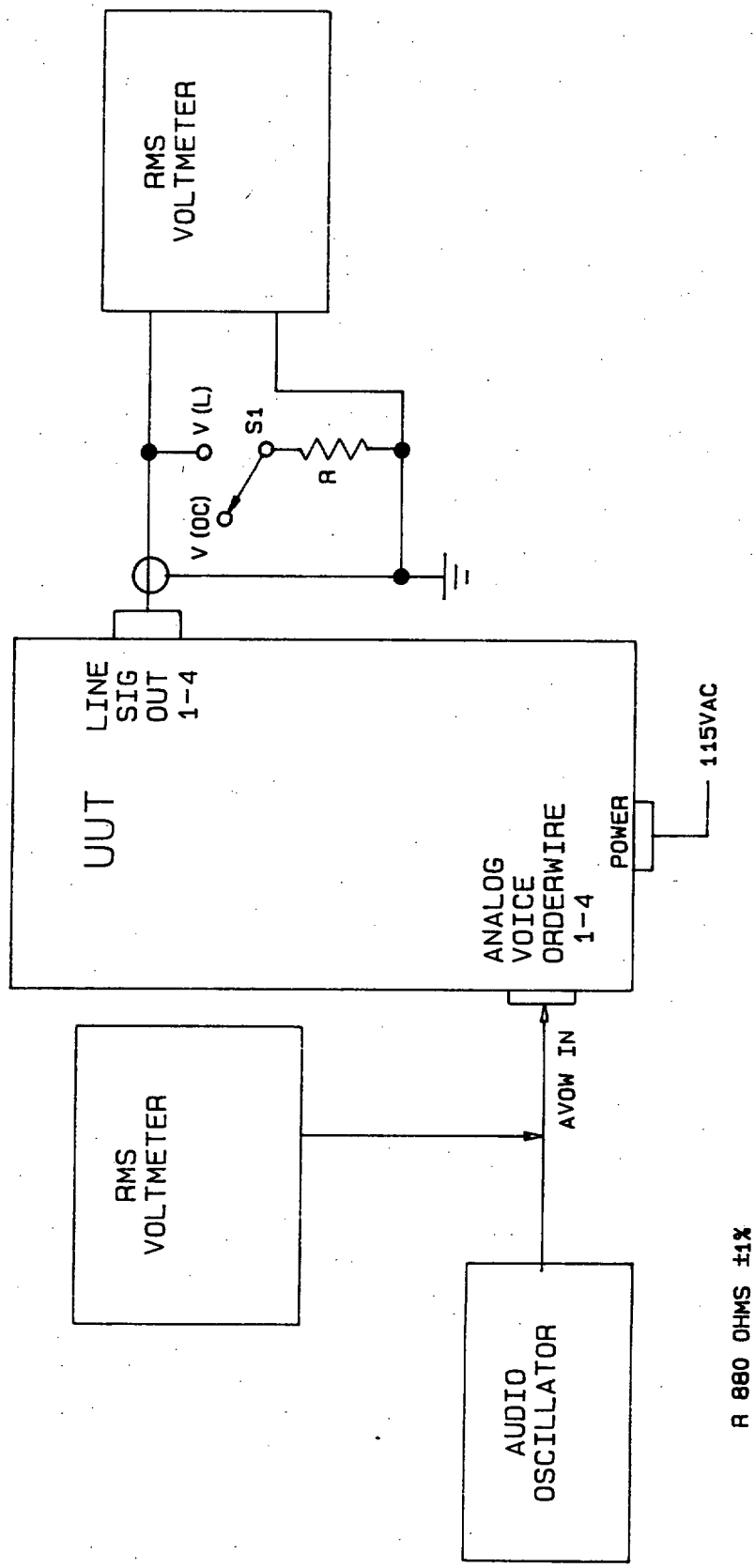


FIGURE 27. Dipulse and bipolar maintenance orderwire cable-side output impedance.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

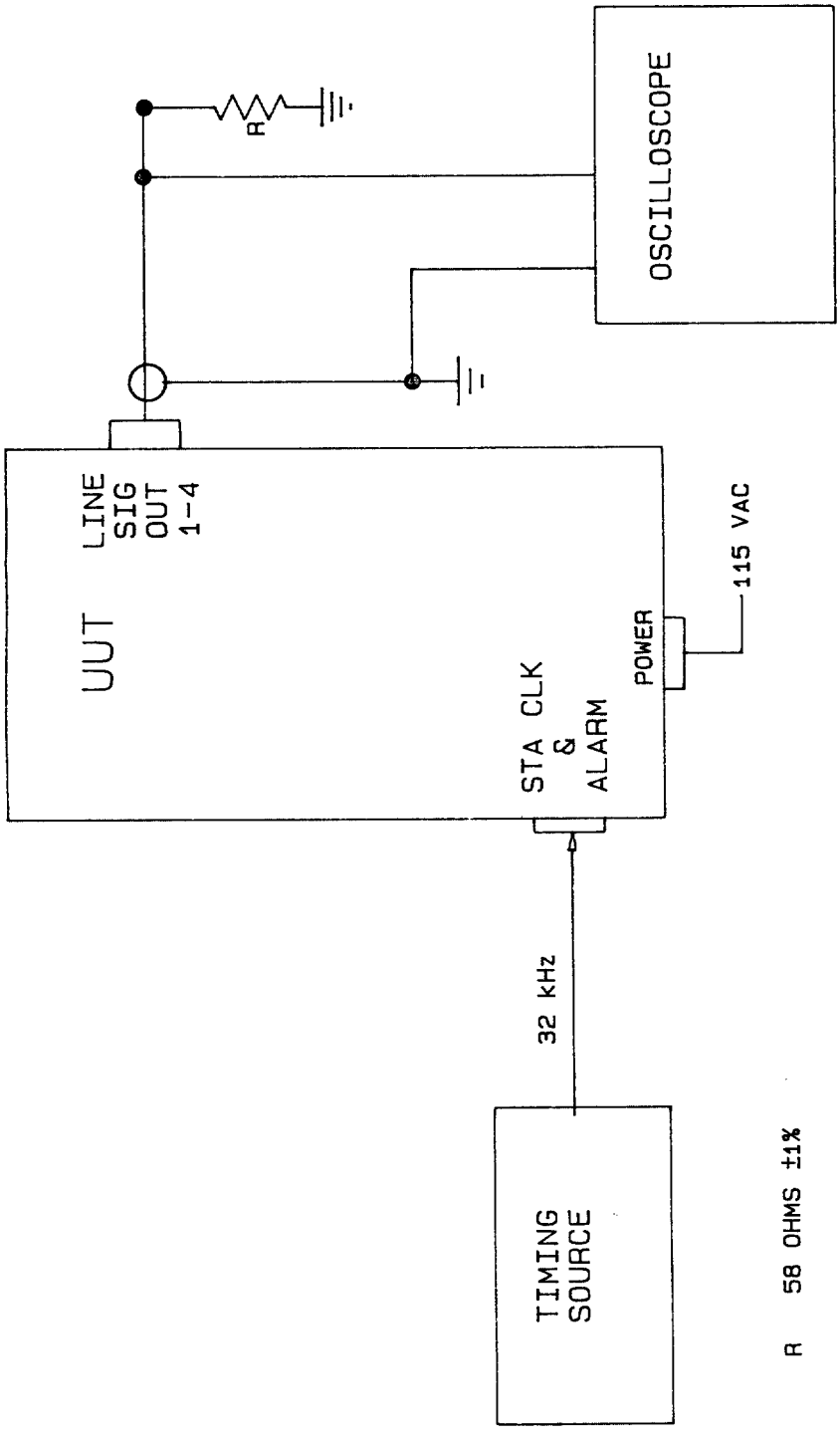


FIGURE 28. Combined digital orderwire cable-side output.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

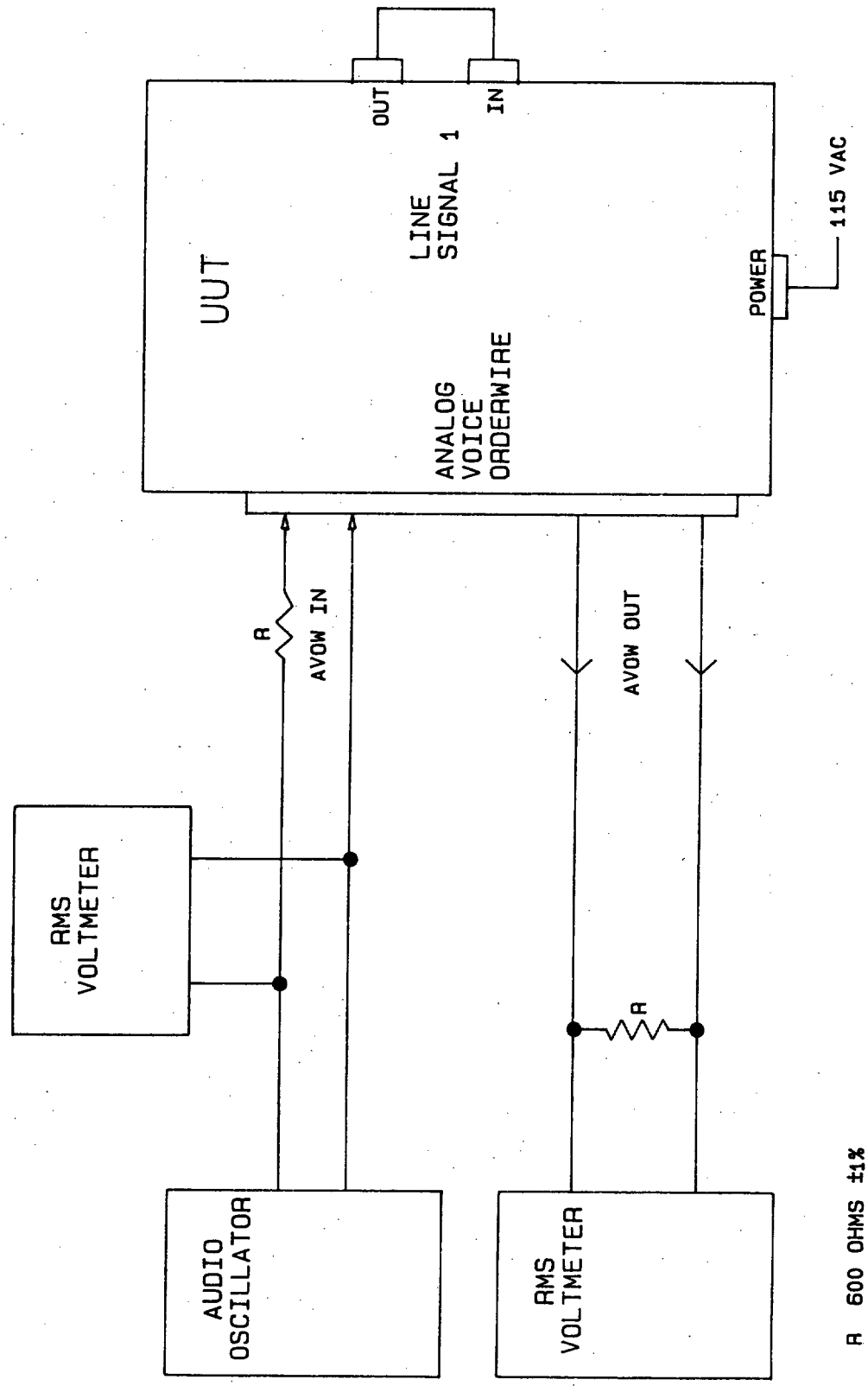
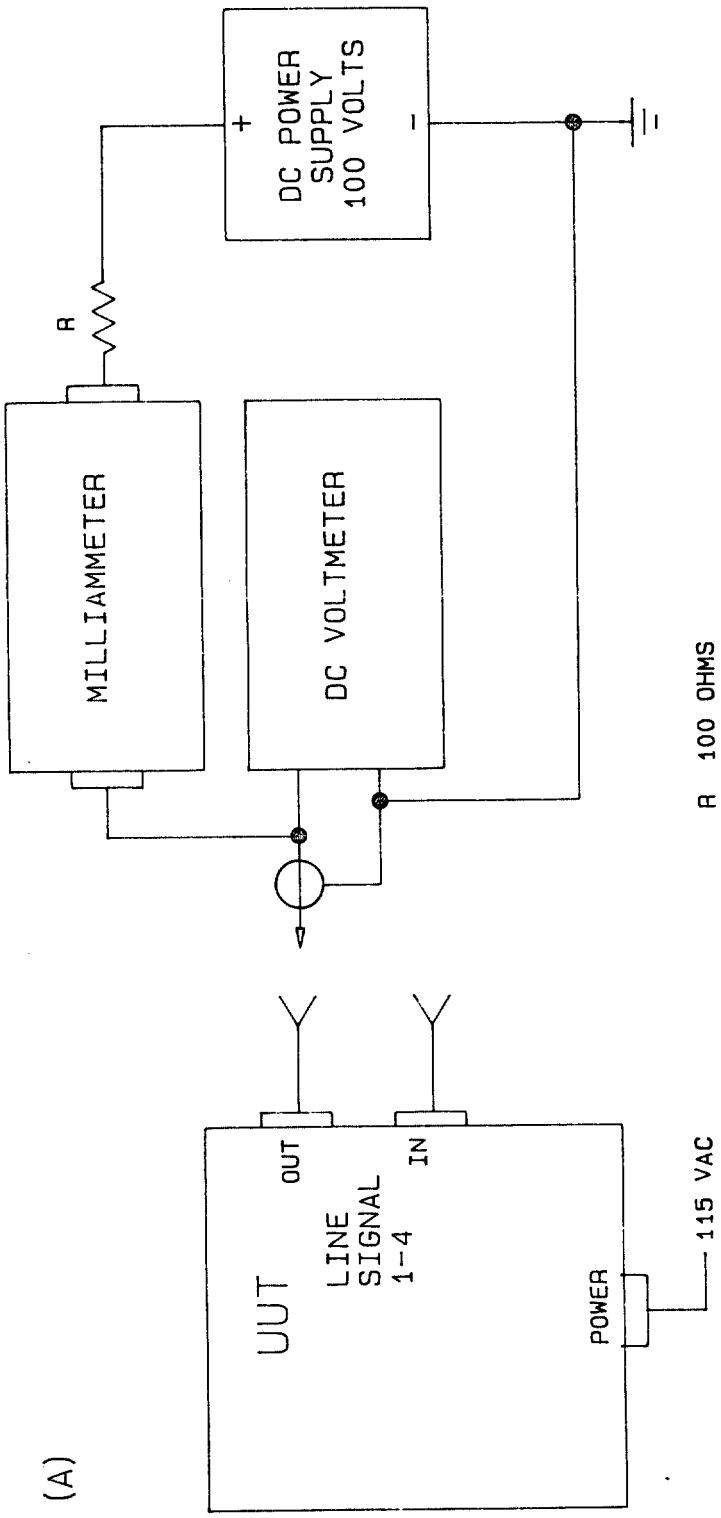


FIGURE 29. Frequency response.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.



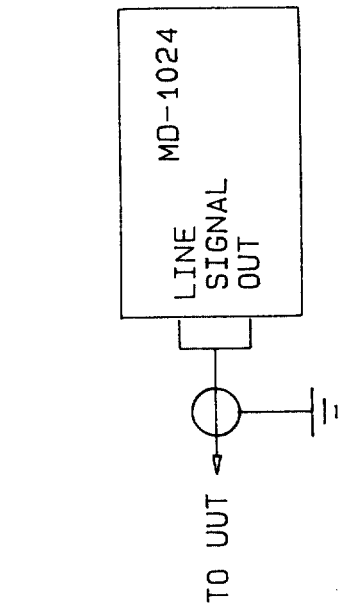


FIGURE 30. Power loop and current protection.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

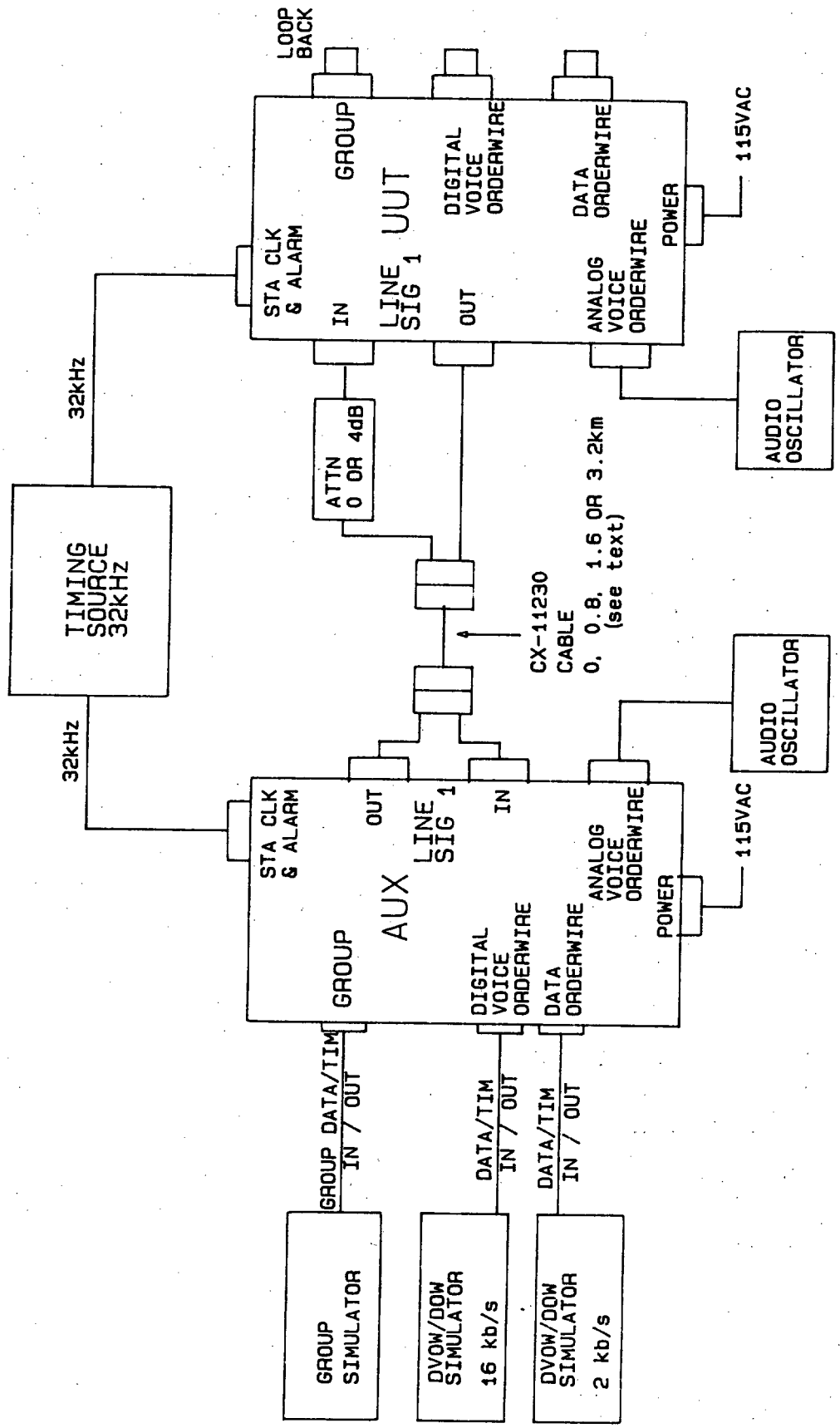


FIGURE 31. Full duplex signal detection; dynamic range; timing source.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

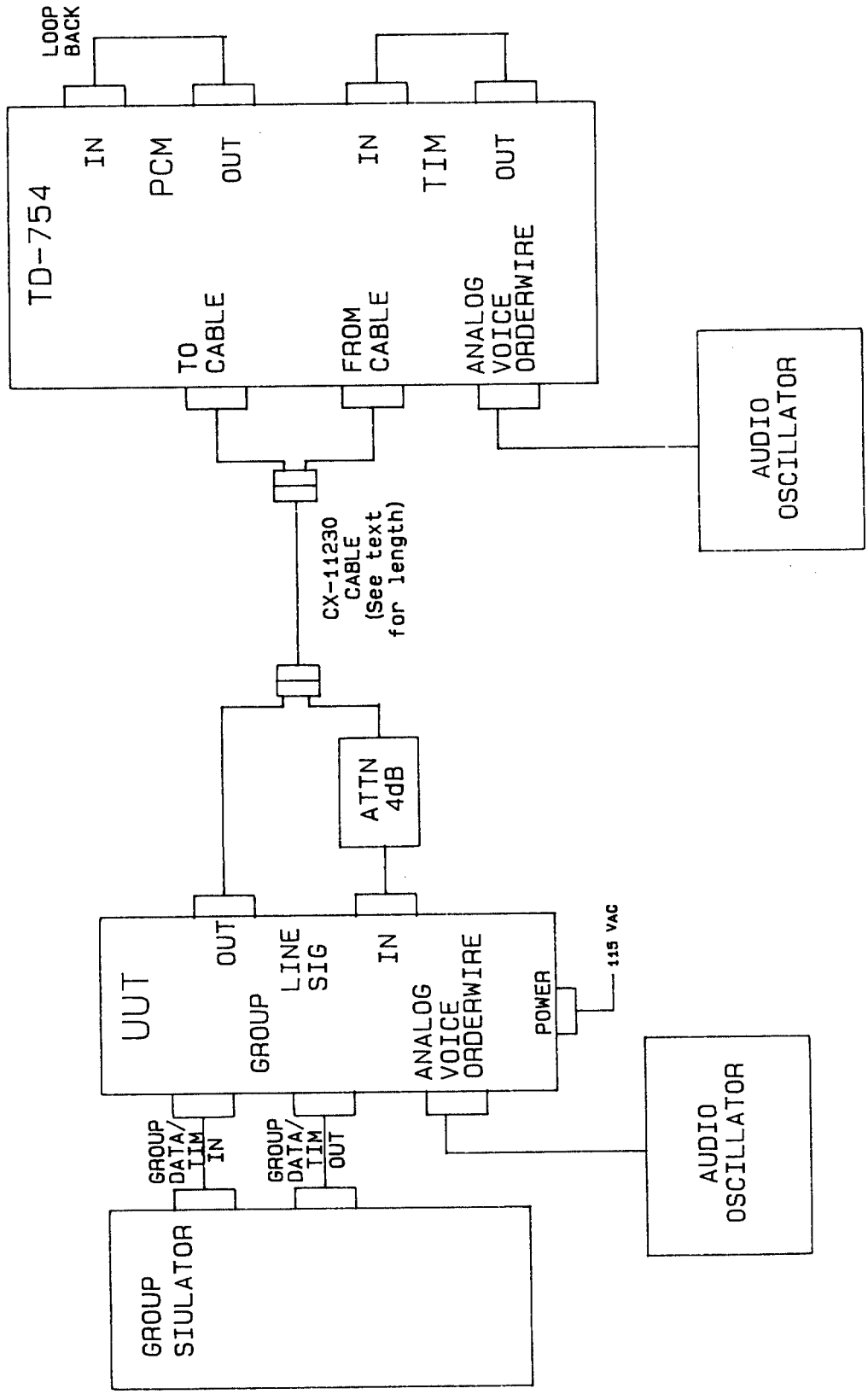


FIGURE 32. Dipulse full-duplex signal detection and dynamic range.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

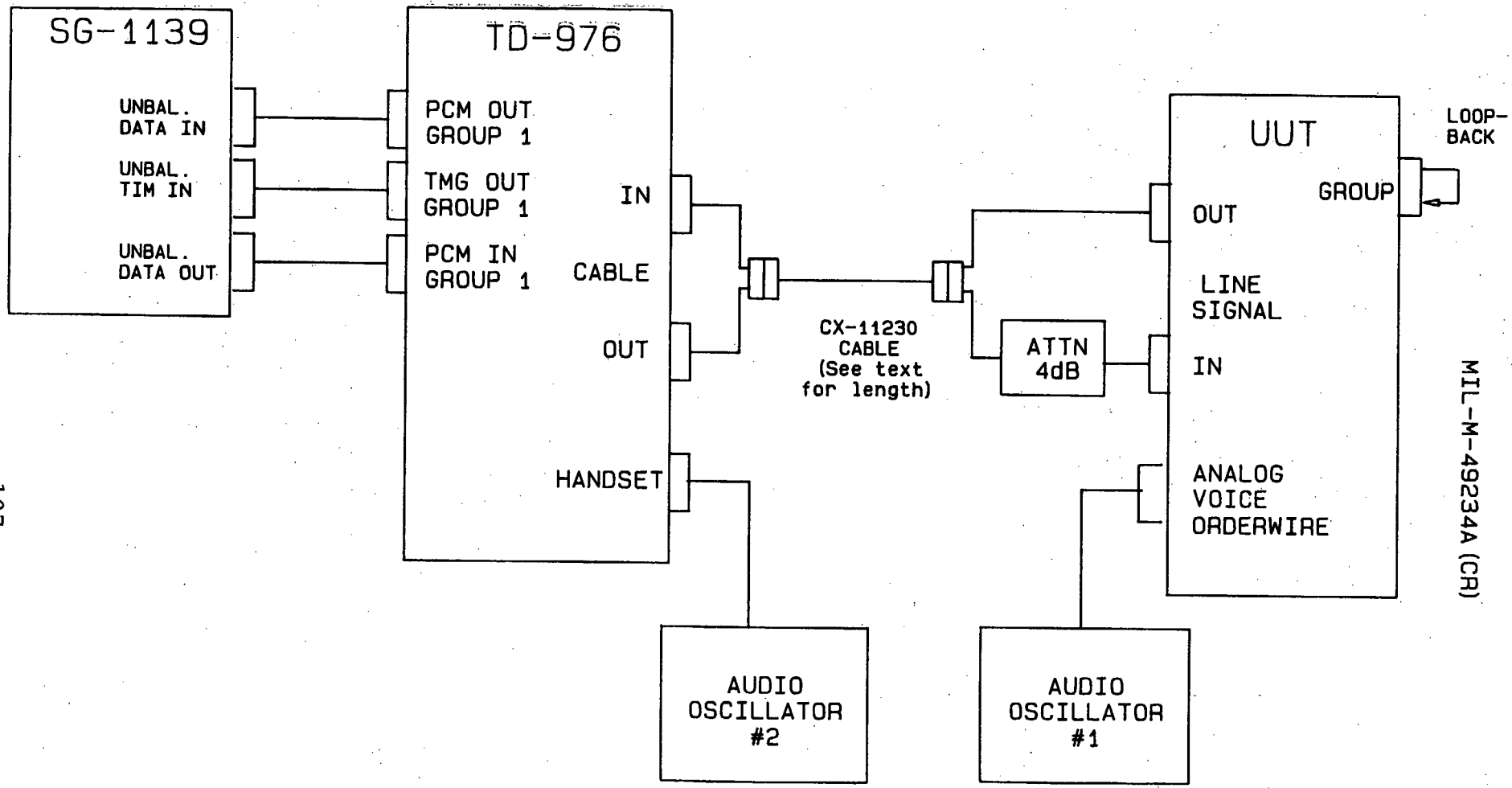


FIGURE 33. Bipolar full-duplex signal detection; dynamic range; timing source.

Source: https://dsist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

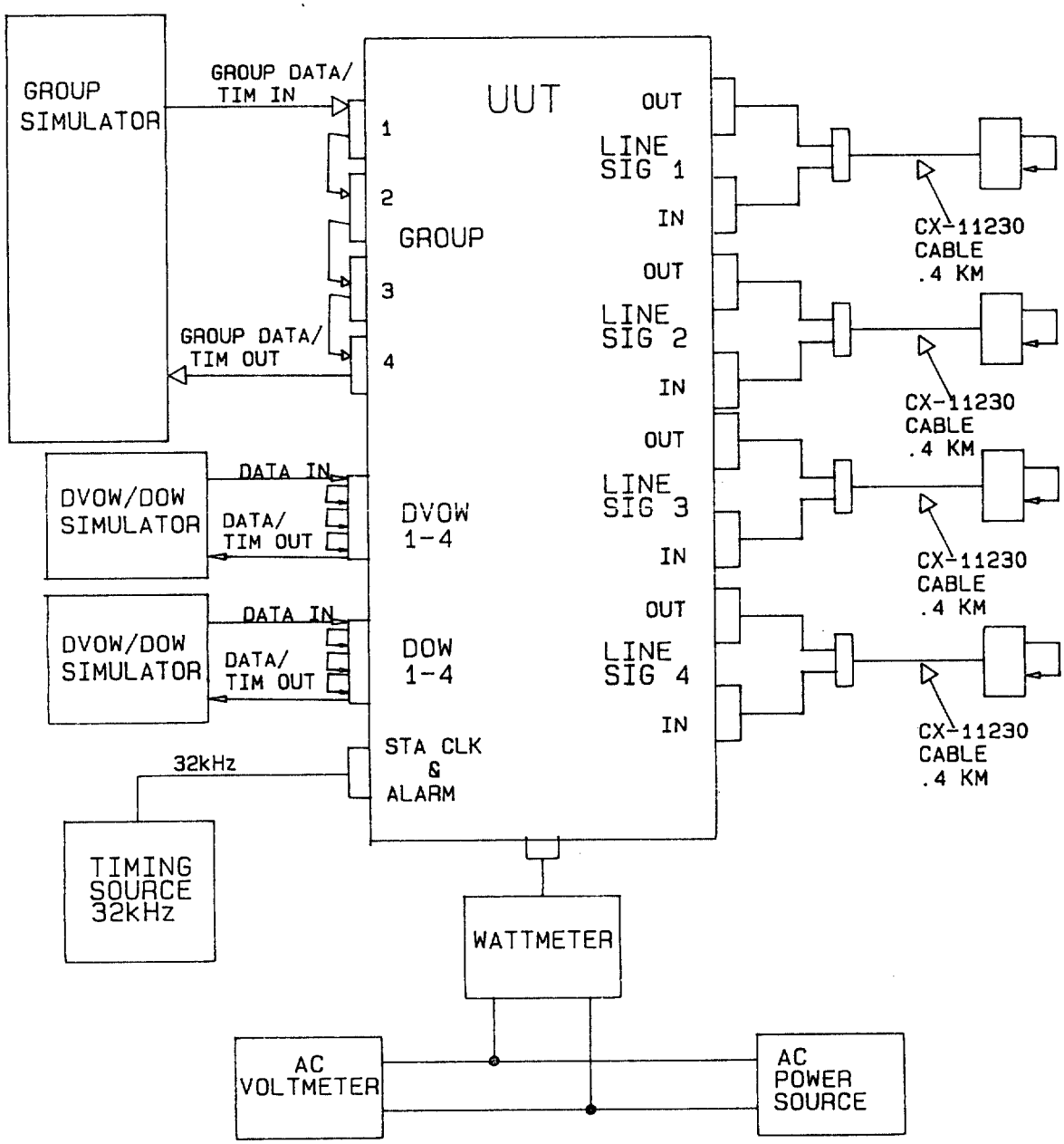


FIGURE 34. Prime power variation.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

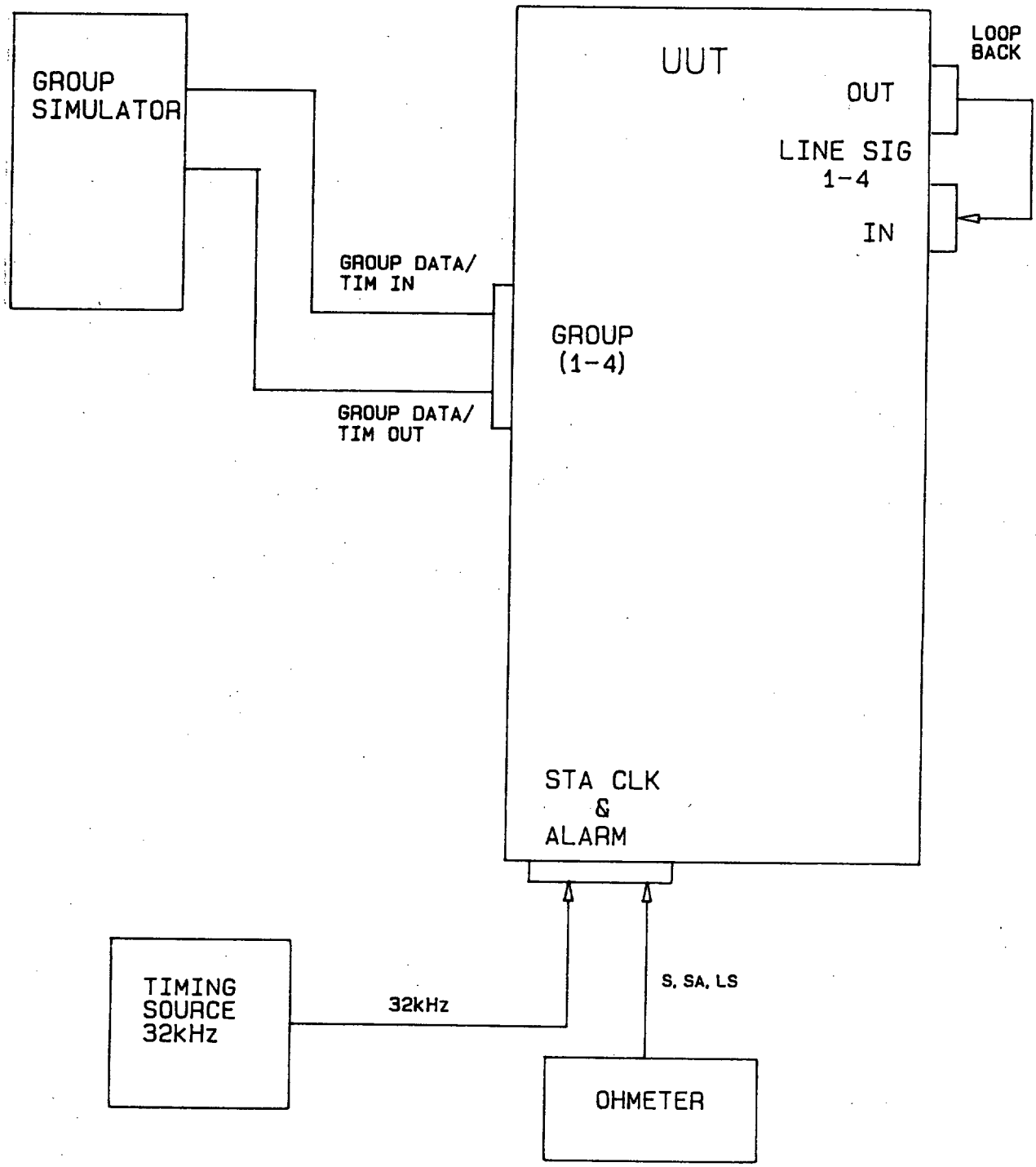
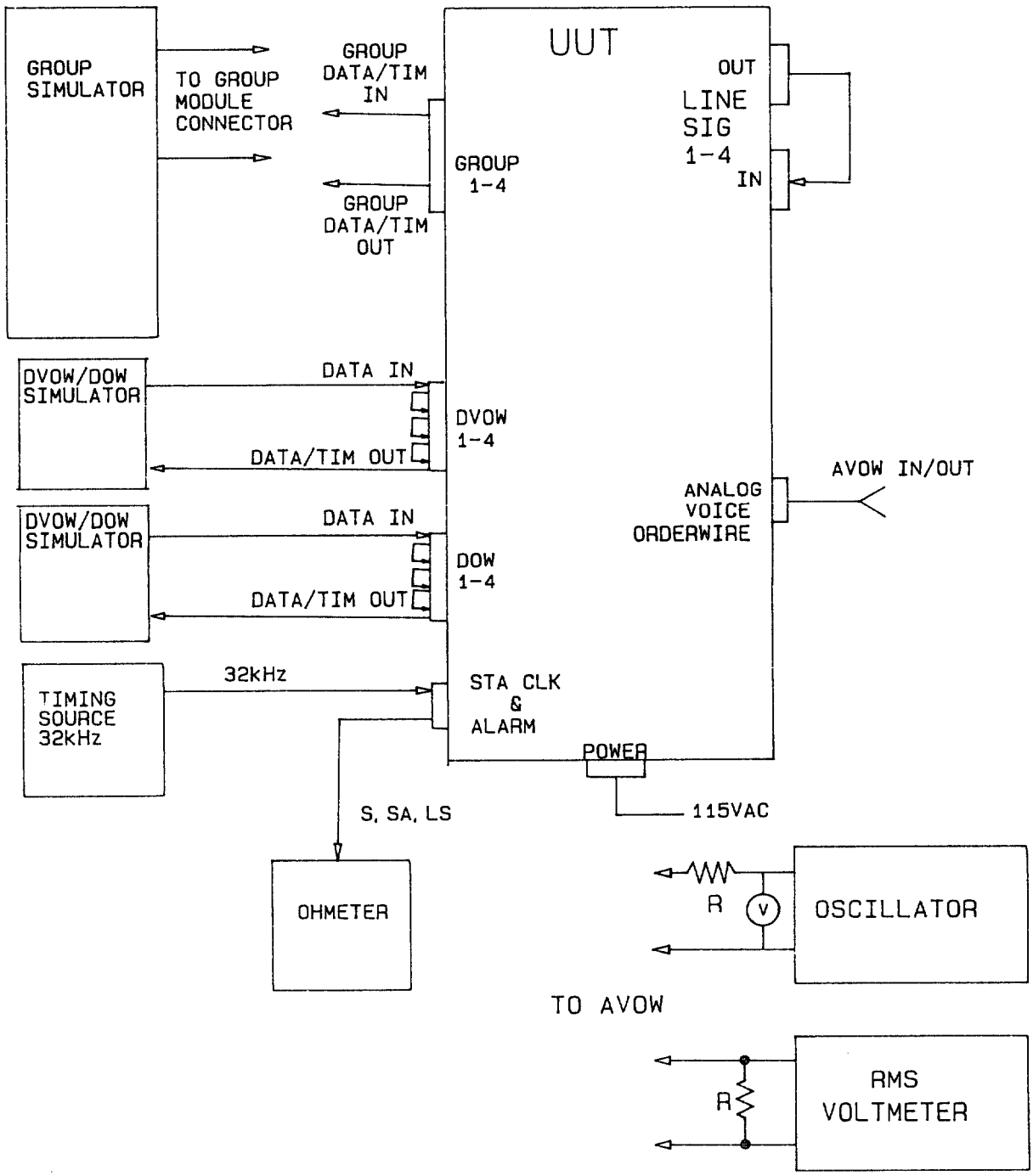


FIGURE 35. BITE and alarm.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.



R 600 OHMS ± 1%

FIGURE 36. Operational tests.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

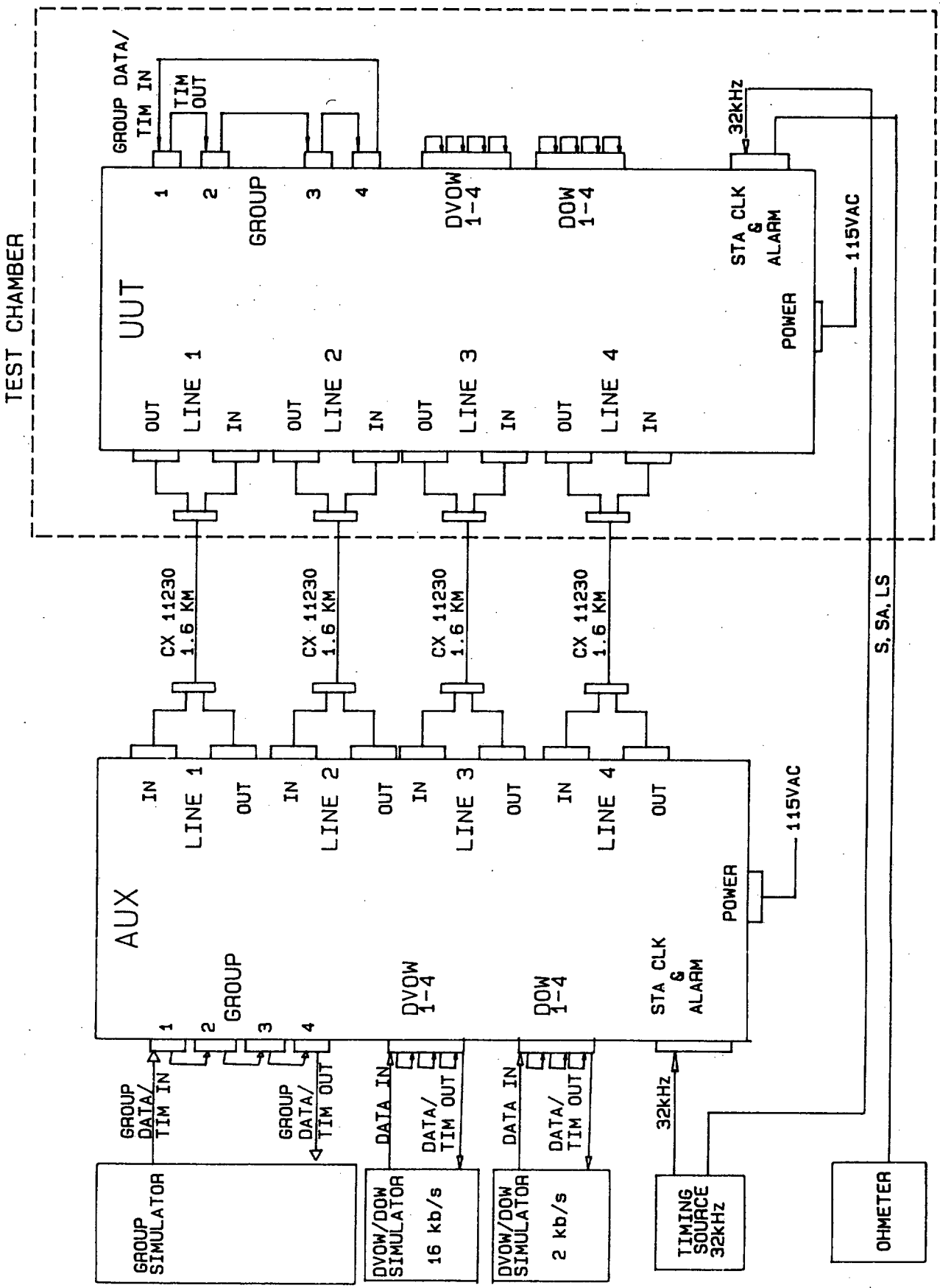


FIGURE 37. Environmental signal detection, diphase configuration.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

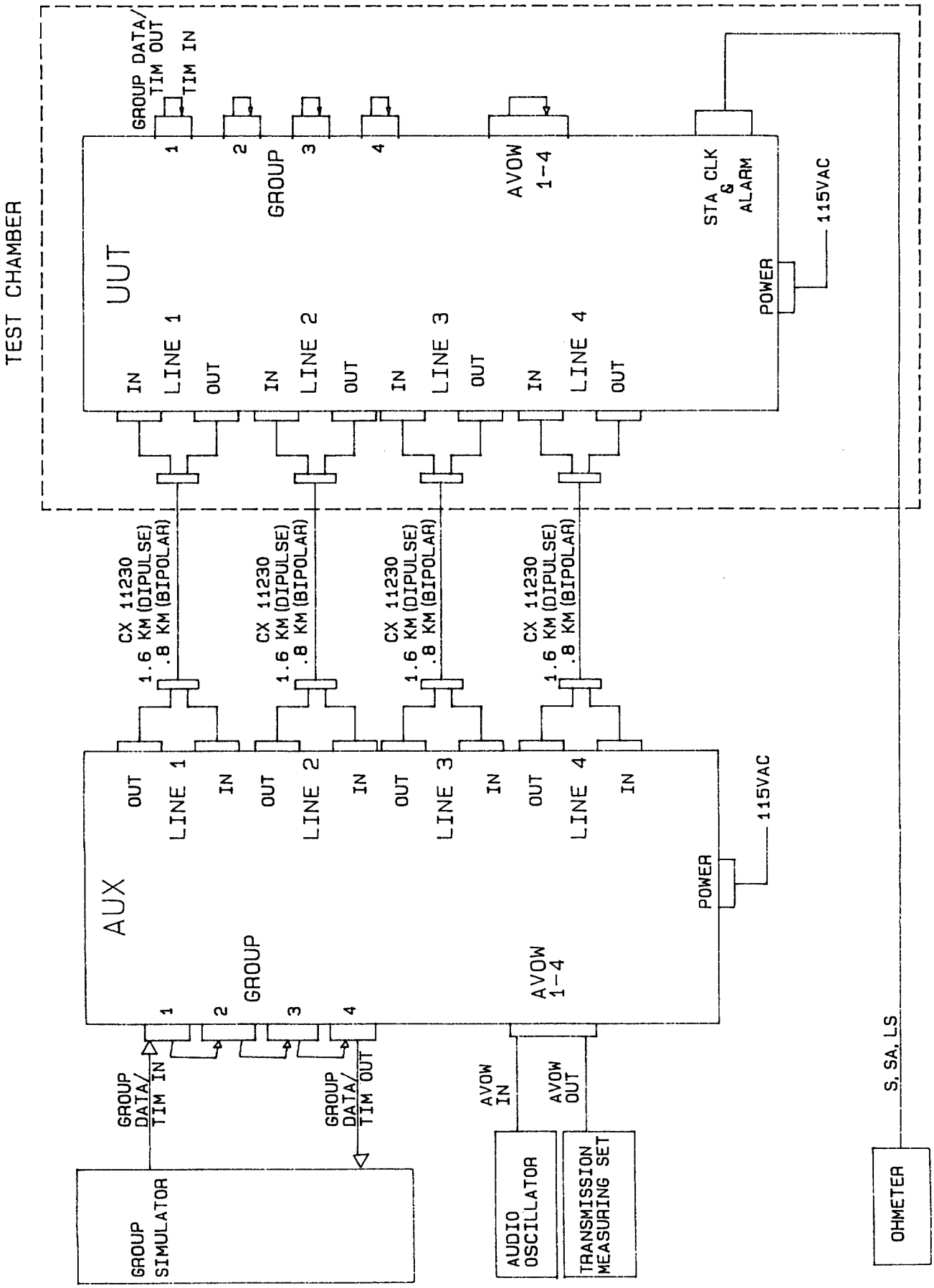


FIGURE 38. Environmental signal detection, dipulse and bipolar configuration.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:59Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-49034A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 82 MAR PREVIOUS EDITION IS OBSOLETE. HISA-FM 149-85

Source: https://assist.dla.mil -- Downloaded: 2026-01-26T05:39Z
Check the source to verify that this is the current version before use.